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Paper sheet handling apparatus

Abstract

A paper sheet handling apparatus includes an accumulation and storage unit that accumulates and stores paper sheets conveyed by the conveyance unit while aligning the paper sheets, the accumulation and storage unit having an accumulation unit that accumulates the paper sheets so that a first direction which is one of a vertical width direction and a horizontal width direction and a second direction which is the other of the vertical width direction and the horizontal width direction are oriented in predetermined directions, a first alignment unit that is movable between a first retracted position spaced apart from the paper sheets and a first alignment position in contact with one end edge of the paper sheets positioned in the first direction; and a second alignment unit that is movable between a second retracted position spaced apart from the paper sheets and a second alignment position in contact with other end edge of the paper sheets positioned in the first direction, with a control unit that causes the paper sheets to be aligned in the first direction by causing the first alignment unit and the second alignment unit to move from the first retracted position and the second retracted position to the first alignment position and the second alignment position and tap the paper sheets accumulated in the accumulation unit from both sides of the first direction.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a paper sheet handling apparatus.

(2) This application is the U.S. national phase of International Application No.

PCT/JP2021/043994 filed Nov. 30, 2021, which designated the U.S. and claims priority to Japanese Patent Application No. 2020-208350 filed on Dec. 16, 2020, the entire contents of each of which are hereby incorporated herein by reference.

BACKGROUND ART

(3) Conventionally, paper sheet handling apparatuses for handling paper sheets have been known (refer to Patent Document 1, for example).

(4) The paper sheet handling apparatus is provided with a tapping mechanism that, by tapping paper sheets sent out from the paper sheet conveyance path, stores the paper sheets in an accumulated state in the accumulation and storage unit while aligning the paper sheets. The tapping mechanism has an elongated cylindrical tapping rod that can be horizontally rotated by a driving force such as a motor.

(5) The tapping bar, by tapping the paper sheets sent out from the paper sheet conveyance path from one end edge side to the other end edge side located in the horizontal width direction of the paper sheets, can cause the other end edge of the paper sheets to abut against (contact) a fixed wall of the accumulation and storage unit. As a result, the paper sheets can be accumulated in the accumulation and storage unit while being aligned along the fixed wall.

PRIOR ART DOCUMENTS

Patent Document

(6) [Patent Document 1]

(7) Japanese Unexamined Patent Application, First Publication No. H09-12205

SUMMARY OF INVENTION

(8) Problems to be Solved by the Invention

(9) However, the paper sheet handling apparatus described in Patent Document 1 aligns the paper sheets by tapping the paper sheets only from one side (one end edge side in the horizontal width direction of the paper sheets) using the tapping bar. Therefore, in this paper sheet handling apparatus, the operation stroke (movement amount) until the paper sheets abut against the fixed wall of the accumulation and storage unit becomes large. Therefore, there is a disadvantage that considerable time is spent on the alignment work.

(10) The present invention has been made in view of such circumstances. An example of an object of the present invention is to provide a paper sheet handling apparatus capable of aligning paper sheets quickly and reliably.

Means for Solving the Problems

(11) A paper sheet handling apparatus according to a first aspect of the present invention includes a conveyance unit that conveys paper sheets, an accumulation and storage unit that accumulates and

stores paper sheets conveyed by the conveyance unit while aligning the paper sheets, and a control unit that controls the accumulation and storage unit, wherein the accumulation and storage unit has: an accumulation unit that accumulates paper sheets so that a first direction, which is one direction of the vertical width direction and the horizontal width direction, and a second direction, which is the other of the vertical width direction and the horizontal width direction, are oriented in predetermined directions; a first alignment unit that is arranged more to one side in the first direction than the accumulation unit and constituted to be movable between a first retracted position spaced apart from the paper sheets accumulated in the accumulation unit, and a first alignment position that is in contact with one end edge positioned in the first direction of the paper sheets accumulated in the accumulation unit; a second alignment unit that is arranged more to the other side in the first direction than the accumulation unit and constituted to be movable between a second retracted position spaced apart from the paper sheets accumulated in the accumulation unit, and a second alignment position that is in contact with the other end edge positioned in the first direction of the paper sheets accumulated in the accumulation unit, and the control unit causes the paper sheets to be aligned in the first direction by causing the first alignment unit and the second alignment unit to move from the first retracted position and the second retracted position to the first alignment position and the second alignment position and tap the paper sheets accumulated in the accumulation unit from both sides in the first direction.

(12) Advantageous Effects of the Invention

(13) According to the present invention, it is possible to provide a paper sheet handling apparatus that quickly and reliably aligns paper sheets.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view showing the appearance of a banknote handling apparatus (paper sheet handling apparatus) according to an embodiment of the present invention.

(2) FIG. 2 is a side view showing the internal configuration of the banknote handling apparatus shown in FIG. 1.

(3) FIG. 3 is a plan view of a banknote showing one example of the front-back directionality pattern of a banknote handled by the banknote handling apparatus shown in FIG. 1.

(4) FIG. 4 is a plan view of a banknote showing another example of the front-back directionality pattern of a banknote handled by the banknote handling apparatus shown in FIG. 1.

(5) FIG. 5 is a plan view of a banknote showing yet another example of the front-back directionality pattern of a banknote handled by the banknote handling apparatus shown in FIG. 1.

(6) FIG. 6 is a plan view of a banknote showing still yet another example of the front-back directionality pattern of a banknote handled by the banknote handling apparatus shown in FIG. 1.

(7) FIG. 7 is a configuration diagram showing a front-back inverting unit and peripheral mechanism units in the banknote handling apparatus shown in FIG. 2.

(8) FIG. 8 is a perspective view showing a first accumulation and storage unit of the banknote handling apparatus shown in FIG. 2.

(9) FIG. 9 is a perspective view showing a width-direction alignment unit of the first accumulation and storage unit shown in FIG. 8.

(10) FIG. 10 is a plan view showing the width-direction alignment unit shown in FIG. 9.

(11) FIG. 11 is a plan view showing a state in which the first alignment unit and the second alignment unit of the width-direction alignment unit shown in FIG. 10 are positioned at the first alignment position and the second alignment position.

(12) FIG. 12 is a plan view showing a state in which the spacing between the first base unit and the second base unit of the width-direction alignment unit shown in FIG. 11 is widened.

- (13) FIG. 13 is a perspective view showing a third alignment mechanism of the first accumulation and storage unit shown in FIG. 8.
- (14) FIG. 14 is a side view showing the third alignment mechanism and fourth alignment mechanism of the first accumulation and storage unit shown in FIG. 8.
- (15) FIG. 15 is a side view showing the alignment plate of the third alignment mechanism shown in FIG. 13 and the peripheral mechanisms thereof.
- (16) FIG. 16 is a side view showing a state in which the alignment plate shown in FIG. 15 is positioned at a third retracted position.
- (17) FIG. 17 is a plan view showing a state in which the spacing between the third base unit of the third alignment mechanism and the fourth base unit of the fourth alignment mechanism shown in FIG. 14 is widened.
- (18) FIG. 18 is a plan view of the width-direction alignment unit when banknotes are accumulated on the upper base unit of the first accumulation and storage unit shown in FIG. 8, with the banknotes accumulated more toward the first alignment unit side.
- (19) FIG. 19 is a plan view showing a state in which the second alignment unit of the width-direction alignment unit is advanced from the state shown in FIG. 18 to be positioned at the second alignment position.
- (20) FIG. 20 is a plan view showing a state in which the horizontal width of banknotes are aligned by positioning the first alignment unit of the width-direction alignment unit to the first alignment position from the state shown in FIG. 19.
- (21) FIG. 21 is a schematic diagram showing a banknote handling apparatus, which is a modification of the banknote handling apparatus of the present embodiment, in which a first accumulation and storage unit and a second accumulation and storage unit are vertically arranged side by side.

EMBODIMENTS FOR CARRYING OUT THE INVENTION

- (22) An embodiment of a paper sheet handling apparatus according to the present invention will be described below with reference to the drawings. In this embodiment, Japanese banknotes (1,000 yen note, 2,000 yen note, 5,000 yen note, and 10,000 yen note) will be described as examples of paper sheets. However, the paper sheets are not limited to this case, and foreign banknotes such as euro banknotes and dollar bills may be used.
- (23) As shown in FIGS. 1 and 2, a banknote handling apparatus (paper sheet handling apparatus according to an embodiment of the present invention) 1 of the present embodiment is provided with an identification and counting device 2, which is a first paper sheet handling apparatus, and an accumulation and bundling device 3, which is a second paper sheet handling apparatus. The identification and counting device 2 and the accumulation and bundling device 3 are connected side by side on a stand 4 provided on the floor, for example.
- (24) In the present embodiment, the direction in which the identification and counting device 2 and the accumulation and bundling device 3 are adjacent to each other is defined as the left-right direction L1. A direction perpendicular to the vertical direction L2 and the left-right direction L1 is defined as a front-rear direction L3. Further, of the front-rear direction L3, the direction from the banknote handling apparatus 1 toward the operator side is defined as front FW, and the opposite direction is defined as rear BK. Furthermore, left and right are defined from the viewpoint of the banknote handling apparatus 1 viewed from the operator side (front FW side). Therefore, when viewed from the operator side, the identification and counting device 2 side is the right side RH, and the accumulation and bundling device 3 side is the left side LH.
- (25) The identification and counting device 2 identifies and counts banknotes S input from the outside under predetermined conditions to perform classification processing, and conveys and delivers the banknotes S to the accumulation and bundling device 3. The accumulation and bundling device 3 performs front-back inversion of the banknotes S conveyed from the identification and counting device 2 based on the identification information of the banknotes S. As

a result, the accumulation and bundling device **3** accumulates the banknotes **S** in the state of the front-back directionalities thereof being unified, and creates a plurality of small bundles.

(26) The identification and counting device **2** and the accumulation and bundling device **3** will be described in detail below.

(27) [Identification and Counting Device]

(28) The identification and counting device **2** will be described in detail below.

(29) As shown in FIGS. **1** and **2**, the identification and counting device **2** performs a predetermined sorting process on banknotes **S** loaded from outside the device (externally) by an operator.

Specifically, the identification and counting device **2** performs a classification process that identifies the banknotes **S** loaded from the outside of the machine and sorts the banknotes by denomination while counting them by denomination, a rejection process that rejects banknotes **S** other than those of the set denominations, and a front-back identification process that identifies the front and back faces of banknotes **S** of the set denominations. Note that the identification and counting device **2** is also called a base machine.

(30) The identification and counting device **2** has a substantially rectangular device case **10**. On the right side surface **10a** of the device case **10**, a loading unit **11** for loading banknotes **S** from outside the device (outside the device case **10**) and a rejection unit **12** for receiving rejected banknotes **S** from inside the device (inside the device case **10**) are formed.

(31) The loading unit **11** and the rejection unit **12** are formed so as to continuously open from the right side surface **10a** of the device case **10** to the front surface **10b**. Note that the rejection unit **12** is arranged above the loading unit **11**.

(32) The identification and counting device **2** is further provided with an identification and conveyance unit **20**, an identification unit **30**, and an outbound conveyance unit **40**. The identification and conveyance unit **20** conveys the banknotes **S** that have been loaded into the loading unit **11**. The identification unit **30** identifies at least the front and back faces of the banknotes **S** being conveyed by the identification and conveyance unit **20**. The outbound conveyance unit **40** receives the banknotes **S** whose front and back faces have been correctly identified by the identification unit **30** from the identification and conveyance unit **20** and conveys them toward the adjacent accumulation and bundling device **3**.

(33) The identification and conveyance unit **20**, the identification unit **30** and the outbound conveyance unit **40** are provided inside the device case **10**.

(34) The identification and counting device **2** has an operation display unit **50**, for example, of a touch panel type, that enables operation input by an operator and displays various kinds of information. The identification and counting device **2** is further provided with a control unit **51** and a power supply unit **52**. The control unit **51** comprehensively controls the identification and counting device **2**. The power supply unit **52** can be connected to an external power supply, and supplies power to each unit of the identification and counting device **2** and the accumulation and bundling device **3**.

(35) The operation display unit **50** is provided on the front surface **10b** of the device case **10** and performs input/output of various signals, information, and the like with the control unit **51**.

(36) The control unit **51** is composed of a microcomputer on which a CPU or the like is mounted, for example, and includes a storage unit **51a** such as a flash memory. The control unit **51** comprehensively controls not only the identification and counting device **2** but also the entire banknote handling apparatus **1** including the accumulation and bundling device **3**. The storage unit **51a** pre-stores, for example, a program or a table for causing the banknote handling apparatus **1** to execute various kinds of arithmetic processing. Furthermore, the storage unit **51a** can store master data that serves as a reference for identification, data of identification and counting results, and the like.

(37) The banknotes **S** are loaded into the loading unit **11** from the outside by the operator. At this time, the banknotes **S** are aligned in a predetermined direction and set on the bottom surface **11a** of

the loading unit **11** in a vertically accumulated state. Specifically, as shown in FIG. **1**, the banknotes **S** are set on the bottom surface **11a** of the loading unit **11** with the long sides of the banknotes **S**, that is, the width direction, aligned with the front-rear direction **L3** of the device, and the short sides, that is, the lengthwise direction, of the banknotes **S** aligned with the left-right direction **L1** of the device.

(38) In particular, when the banknotes **S** are set in the loading unit **11**, in relation to the front and back faces, they are set in a state in which four front and back patterns are mixed as shown in FIGS. **3** to **6**.

(39) As the first front-back pattern, as shown in FIG. **3**, the surface on which a specific portrait **M** is clearly shown faces upward, and the portrait **M** faces the left side **LH** of the apparatus. In the present embodiment, this pattern is referred to as a banknote **S** in the first front orientation **N1**.

(40) As the second front-back pattern, as shown in FIG. **4**, the surface on which a specific portrait **M** is clearly shown faces downward, and the portrait **M** faces the right side **RH** of the apparatus. In the present embodiment, this pattern is referred to as a banknote **S** in the first back orientation **N2**.

(41) As the third front-back pattern, as shown in FIG. **5**, the surface on which a specific portrait **M** is clearly shown faces upward, and the portrait **M** faces the right side **RH** of the apparatus. In the present embodiment, this pattern is referred to as a banknote **S** in the second front orientation **N3**.

(42) As the fourth front-back pattern, as shown in FIG. **6**, the surface on which a specific portrait **M** is clearly shown faces downward, and the portrait **M** faces the left side **LH** of the apparatus. In the present embodiment, this pattern is referred to as a banknote **S** in the second back orientation **N4**.

(43) In this way, the banknotes **S** are set in an accumulated state on the bottom surface **11a** of the loading unit **11** shown in FIGS. **1** and **2** in the state of the above-described four front and back patterns being mixed. The loading unit **11** has a bottom surface **11a**, a wall surface **11b**, and a pair of side surfaces **11c**. The bottom surface **11a** is inclined slightly downward to the left with respect to the floor surface. The wall surface **11b** extends upward from the left end of the bottom surface **11a**. The pair of side surfaces **11c** are integrally connected to the bottom surface **11a** and the wall surface **11b** and face each other in the front-rear direction **L3**.

(44) A bill press **13** that moves up and down along the wall surface **11b** is provided on the wall surface **11b**. The bill press **13** presses down the banknotes **S** placed on the bottom surface **11a** from above, and can hold the banknotes **S** stably by sandwiching the banknotes **S** with the bottom surface **11a**.

(45) As shown in FIG. **2**, the loading unit **11** is provided with a feed-out roller **14** and a take-in roller **15**. Out of the banknotes **S** set on the bottom surface **11a**, the feed-out roller **14** feeds out the lowest banknote **S** one by one toward the left side **LH** of the apparatus. The take-in roller **15** takes in the banknotes **S** fed out by the feed-out roller **14** into the device and transfers the banknotes **S** to the identification and conveyance unit **20**. Among the banknotes **S** set in the loading unit **11**, only the lowest banknote **S** can be appropriately separated one by one by the feed-out roller **14** and the take-in roller **15** and transferred to the identification and conveyance unit **20**.

(46) The banknotes **S** transferred from the loading unit **11** to the identification and conveyance unit **20** thereafter move through the entire conveyance unit in the banknote handling apparatus **1** maintaining the orientation in which the short sides of the banknotes **S**, that is, the vertical width direction, are aligned with the left-right direction **L1** or the vertical direction **L2** of the apparatus. Therefore, the vertical width direction of the banknotes **S** coincides with the left-right direction **L1** or vertical direction **L2** of the apparatus, as well as with the direction of conveyance.

(47) The identification and conveyance unit **20** is provided with a first identification and conveyance unit **21** extending from the loading unit **11** toward the left side **LH**. Moreover, the identification and conveyance unit **20** is provided with a second identification and conveyance unit **22** extending upward from the left end unit of the first identification and conveyance unit **21**. Thereby, the banknotes **S** transferred from the loading unit **11** to the identification and conveyance unit **20** are conveyed toward the left side **LH** by the first identification and conveyance unit **21**.

Subsequently, the banknotes S are conveyed by the second identification and conveyance unit 22 with an upward orientation. Further, near the take-in roller 15 of the first identification and conveyance unit 21, a detection unit 23 is installed for detecting whether or not a banknote S has been taken in.

(48) The outbound conveyance unit 40 is provided to extend from the upper end unit of the second identification and conveyance unit 22 toward the left side LH. The outbound conveyance unit 40 conveys the banknotes S toward a coupled conveyance unit 70 of the accumulation and bundling device 3, which will be described later. Moreover, a rejection conveyance unit 41 extending toward the right side RH is connected to the upper end unit of the second identification and conveyance unit 22. The rejection conveyance unit 41 is connected to the rejection unit 12. As a result, the outbound conveyance unit 40 and the rejection conveyance unit 41 are branched and connected to the upper end unit of the second identification and conveyance unit 22.

(49) A sorting unit 42 is provided at the upper end of the second identification and conveyance unit 22. The sorting unit 42 sorts the banknotes S conveyed by the second identification and conveyance unit 22 to either the outbound conveyance unit 40 or the rejection conveyance unit 41. In other words, the sorting unit 42 plays a role of switching the conveyance destination of the banknotes S from the second identification and conveyance unit 22 to the outbound conveyance unit 40 or the rejection conveyance unit 41. The sorting unit 42 switches the conveyance destination of the banknotes S to the outbound conveyance unit 40 or the rejection conveyance unit 41 on the basis of the identification result of the identification unit 30.

(50) The identification unit 30 is provided in the second identification and conveyance unit 22 and performs identification and counting of the banknotes S.

(51) For example, the identification unit 30 identifies the type such as the denomination, and identifies the front-back pattern of each banknote S as one of the four front-back patterns described above (first front orientation N1, first back orientation N2, second front orientation N3, second back orientation N4).

(52) Note that the identification unit 30 outputs the identification information (identification result) of the banknote S to the control unit 51.

(53) Based on the identification information output from the identification unit 30, the control unit 51 determines whether the banknote S conveyed from the loading unit 11 can be accepted and whether it is of a preset denomination. When the conveyed banknote S is acceptable and is of a preset denomination, the control unit 51 has the banknote S conveyed from the second identification and conveyance unit 22 through the outbound conveyance unit 40 to the accumulation and bundling device 3.

(54) Further, the control unit 51, upon determining that the banknote S conveyed from the loading unit 11 cannot be accepted based on the identification information output from the identification unit 30, determines the banknote S to be a banknote S to be rejected. In this case, the control unit 51 causes the banknote S to be conveyed from the second identification and conveyance unit 22 to the rejection unit 12 via the rejection conveyance unit 41.

(55) Furthermore, upon determining, based on the identification information output from the identification unit 30, that the banknote S conveyed from the loading unit 11 is acceptable and has a denomination other than the preset denomination, for example, a 1,000-yen note different from the preset 10,000-yen note, the control unit 51 determines the conveyed banknote S to be a banknote S to be rejected. In this case, the control unit 51 causes the conveyed banknote S to be conveyed from the second identification and conveyance unit 22 to the rejection unit 12 via the rejection conveyance unit 41.

(56) As shown in FIGS. 1 and 2, in the rejection unit 12, the banknotes S are fed out from the right end of the rejection conveyance unit 41, with the banknotes S being accumulated so as to be removable to the outside. In other words, the rejection unit 12 is an ejection port for ejecting banknotes S conveyed by the rejection conveyance unit 41 out of the machine.

(57) The rejection unit **12** has a bottom surface **12a**, a wall surface **12b**, and a pair of side surfaces **12c**. The bottom surface **12a** is inclined slightly downward to the left with respect to the floor surface. The wall surface **12b** extends upward from the left end of the bottom surface **12a**. The pair of side surfaces **12c** are integrally connected to the bottom surface **12a** and the wall surface **12b** and face each other in the front-rear direction **L3**.

(58) Further, the rejection unit **12** has an impeller **43** that causes the banknotes **S** fed out from the rejection conveyance unit **41** to fall toward the bottom surface **12a** while being brought into contact with the wall surface **12b**. Thereby, the rejection unit **12** can place the banknotes **S** fed out from the rejection conveyance unit **41** on the bottom surface **12a** while stably accumulating them thereon.

(59) [Accumulation and Bundling Device]

(60) The accumulation and bundling device **3** will be described in detail below.

(61) As shown in FIGS. **1** and **2**, the accumulation and bundling device **3** is arranged on the left side **LH** of the identification and counting device **2** described above, and is equipped with a substantially rectangular device case **60**. The device case **10** of the accumulation and bundling device **3** and the device case **60** of the identification and counting device **2** are integrally combined by a known method such as screw connection or pin connection.

(62) However, the invention is not limited to the case where the separate device cases **10** and **60** are respectively provided. For example, the banknote handling apparatus **1** may have one common apparatus case, and the identification and counting device **2** and the accumulation and bundling device **3** may be arranged in this common apparatus case.

(63) The accumulation and bundling device **3** performs front-back inversion of the banknotes **S** of a specific denomination conveyed by the outbound conveyance unit **40** of the identification and counting device **2** based on the identification information of the banknotes **S**. As a result, the front-back directionalities of the banknotes **S** are unified, and the banknotes **S** with the front-back directionalities thus unified are bundled to produce a small bundle.

(64) As shown in FIG. **2**, the accumulation and bundling device **3** has a coupled conveyance unit (conveyance unit according to the present invention) **70** that receives from the outbound conveyance unit **40** banknotes **S** whose front and back faces have been identified and conveys them. Further, the accumulation and bundling device **3** is provided with a front-back inverting unit **80** that inverts the banknotes **S** being conveyed by the coupled conveyance unit **70** based on the front-back identification information detected by the identification unit **30**. The coupled conveyance unit **70** and the front-back inverting unit **80** are provided inside the device case **60**.

(65) (Coupled Conveyance Unit)

(66) The coupled conveyance unit **70** is arranged on the upper surface side in the device case **60** and is arranged so as to linearly extend along the left-right direction **L1**. The coupled conveyance unit **70** conveys the banknotes **S** from the upstream right end to the downstream left end. The right end of the coupled conveyance unit **70** is connected to the outbound conveyance unit **40** of the identification and counting device **2**. On the other hand, the left end of the coupled conveyance unit **70** opens to the left side surface of the device case **60**, and for example can be used as an optional exit **71** that can be used for multiple purposes, such as conveying banknotes **S** to an evacuation pocket unit capable of storing banknotes that is installed on the left side surface of the device case **60** of the accumulation and bundling device **3**.

(67) As a result, the coupled conveyance unit **70** conveys the banknotes **S** received from the outbound conveyance unit **40** toward the left side **LH**. At this time, the banknotes **S** conveyed by the coupled conveyance unit **70** are a mixture of the above-described four front-back patterns. That is, banknotes in the first front orientation **N1**, banknotes in the first back orientation **N2**, banknotes in the second front orientation **N3**, and banknotes in the second back orientation **N4** are conveyed by the coupled conveyance unit **70**.

(68) (Front-Back Inverting Unit)

(69) As shown in FIGS. **2** and **7**, two front-back inverting units **80** are provided coupled to the

coupled conveyance unit **70**.

(70) Specifically, the front-back inverting units **80** are arranged side by side in the left-right direction **L1** and are provided at the same height position within the device case. Therefore, the two front-back inverting units **80** are arranged adjacent to each other along the conveyance direction of the banknotes **S** conveyed by the coupled conveyance unit **70**.

(71) In the present embodiment, of the two front-back inverting units **80**, the front-back inverting unit **80** on the right side **RH** positioned closer to the identification and counting device **2** is referred to as a first front-back inverting unit **81**, and the front-back inverting unit **80** positioned on the left side **LH** is referred to as a second front-back inverting unit **82**.

(72) (First Front-Back Inverting Unit)

(73) The first front-back inverting unit **81** is provided so as to intersect the coupled conveyance unit **70**. The front-back inverting unit **81** is provided with a branching conveyance unit **90**, a non-inverting unit **91**, and an inverting unit **92**. The branching conveyance unit **90** is coupled to the coupled conveyance unit **70**. The non-inverting unit **91** inverts the banknotes **S** an even number of times based on the identification information detected by the identification unit **30**, and so discharges the banknotes **S** with the same front-back directionality as before the front-back inversion. The non-inverting unit **92** inverts the banknotes **S** an odd number of times based on the identification information detected by the identification unit **30**, and so discharges the banknotes **S** with a different front-back directionality than before the front-back inversion.

(74) The branching conveyance unit **90** is arranged below the coupled conveyance unit **70** and is formed to extend in the vertical direction **L2**. The upper end unit of the branching conveyance unit **90** is connected to the coupled conveyance unit **70** from below so as to intersect the coupled conveyance unit **70** substantially at a right angle.

(75) At the connection unit between the upper end unit of the branching conveyance unit **90** and the coupled conveyance unit **70** is provided a branch sorting unit **93** that sorts the banknotes **S** having two front-back patterns among the banknotes **S** conveyed from the coupled conveyance unit **70** to the branching conveyance unit **90**, and flows the remaining banknotes **S** to the second front-back inverting unit **82** side positioned downstream of the branching conveyance unit **90**.

(76) Specifically, the branch sorting unit **93** sorts only the banknotes in the first front orientation **N1** and the banknotes in the first back orientation **N2** among the banknotes **S** conveyed by the coupled conveyance unit **70** to the branching conveyance unit **90**. The branch sorting unit **93** passes the remaining banknotes **S** in the second front orientation **N3** and banknotes **S** in the second back orientation **N4**, sorting them toward the second front-back inverting unit **82** side.

(77) The operation of the branch sorting unit **93** is controlled by the control unit **51**. Also, the control unit **51** controls the branch sorting unit **93** based on the identification information.

(78) The non-inverting unit **91** is arranged below the coupled conveyance unit **70** and on the right side **RH** of the branching conveyance unit **90**. In contrast, the inverting unit **92** is arranged below the coupled conveyance unit **70** and on the left side **LH** of the branching conveyance unit **90**. Thereby, the non-inverting unit **91** and the inverting unit **92** are arranged side by side in the left-right direction **L1** with the branching conveyance unit **90** interposed therebetween.

(79) The non-inverting unit **91** is connected to the branching conveyance unit **90**, and is formed to turn 180 degrees after heading from the branching conveyance unit **90** to the right end unit side that is the upstream side of the coupled conveyance unit **70**, to head to the left end unit side that is the downstream side of the coupled conveyance unit **70**.

(80) Specifically, the non-inverting unit **91** includes a first non-inverting unit **91a**, a second non-inverting unit **91b**, and a third non-inverting unit **91c**. The first non-inverting unit **91a** is connected to a portion located between the upper end portion and the lower end portion of the branching conveyance unit **90** and extends toward the upstream side of the coupled conveyance unit **70** along the left-right direction **L1**. The second non-inverting unit **91b** is connected to the right end side of the first non-inverting unit **91a** and extends downward along the vertical direction **L2**. The third

non-inverting unit **91c** is connected to the lower end side of the second non-inverting unit **91b** and extends toward the downstream side of the coupled conveyance unit **70** along the left-right direction **L1**. Thereby, the non-inverting unit **91** as a whole is formed in a U shape that bulges toward the upstream side of the coupled conveyance unit **70**.

(81) However, the overall shape of the non-inverting unit **91** is not limited to a U shape, and may be formed to bulge toward the upstream side of the coupled conveyance unit **70** in various curved shapes such as a semicircular shape, an arc shape, a C shape (katakana “ko” shape), or the like.

(82) Since the non-inverting unit **91** is configured as described above, while the banknote S is being conveyed from the coupled conveyance unit **70** to the first non-inverting unit **91a** of the non-inverting unit **91** via the branching conveyance unit **90**, the banknote S can be front-back inverted once. Subsequently, while the banknote S is conveyed from the first non-inverting unit **91a** to the third non-inverting unit **91c** via the non-inverting unit **91b**, the banknote S can be front-back inverted again.

(83) As a result, the non-inverting unit **91** front-back inverts the banknote S twice (an even number of times) in total, and so can discharge the banknote S from the left end unit side of the third non-inverting unit **91c** with the same front-back directionality as the banknote S when conveyed by the coupled conveyance unit **70**.

(84) The inverting unit **92** is connected to the branching conveyance unit **90**, and is formed to turn 180 degrees after heading from the branching conveyance unit **90** to the left end portion side that is the downstream side of the coupled conveyance unit **70**, to head to the right end portion side that is the upstream side of the coupled conveyance unit **70**.

(85) Specifically, the inverting unit **92** includes a first inverting unit **92a**, a second inverting unit **92b**, and a third inverting unit **92c**. The first inverting unit **92a** is connected to the lower end portion of the branching conveyance unit **90** and extends toward the downstream side of the coupled conveyance unit **70** along the left-right direction **L1**. The second inverting unit **92b** is connected to the left end side of the first inverting unit **92a** and extends downward along the vertical direction **L2**. The third inverting unit **92c** is connected to the lower end side of the second inverting unit **92b** and extends toward the upstream side of the coupled conveyance unit **70** along the left-right direction **L1**. Thereby, the inverting unit **92** as a whole is formed in a U shape that bulges toward the downstream side of the coupled conveyance unit **70**.

(86) However, the overall shape of the non-inverting unit **92** is not limited to a U shape, and may be formed to bulge toward the downstream side of the coupled conveyance unit **70** in various curved shapes such as a semicircular shape, an arc shape, a C shape (katakana “ko” shape), or the like.

(87) Since the inverting unit **92** is configured as described above, while the banknote S is being conveyed from the coupled conveyance unit **70** to the first inverting unit **92a** of the inverting unit **92** via the branching conveyance unit **90**, the banknote S cannot be front-back inverted. After that, while the banknote S is being conveyed from the first inverting unit **92a** to the third inverting unit **92c** via the second inverting unit **92b**, the banknote S can be front-back inverted once.

(88) As a result, the non-inverting unit **92** front-back inverts the banknote S once (an odd number of times) in total, and so can discharge the banknote S from the right end portion side of the third inverting unit **92c** with a different front-back directionality as the banknote S when conveyed by the coupled conveyance unit **70**.

(89) Note that the position of the left end of the third non-inverting unit **91c** in the non-inverting unit **91** (discharge position of the banknote S) and the position of the right end of the third inverting unit **92c** in the inverting unit **92** (discharge position of the banknote S) are positioned at the same height, and are arranged to face each other with a gap in the left-right direction **L1**.

(90) In the present embodiment, since the non-inverting unit **91** is connected to a portion located between the upper end portion and the lower end portion of the branching conveyance unit **90**, and the inverting unit **92** is connected to the lower end portion of the branching conveyance unit **90**, the

connection position between the branching conveyance unit **90** and the non-inverting unit **91** is located higher than the connection position between the branching conveyance unit **90** and the inverting unit **92**.

(91) However, the present invention is not limited to this case, and the connection position between the branching conveyance unit **90** and the inverting unit **92** may be positioned above the connection position between the branching conveyance unit **90** and the non-inverting unit **91**.

(92) At the connection unit between the branching conveyance unit **90** and the non-inverting unit **91**, there is provided a gate unit **94** for switching the conveyance path so as to convey the banknotes S conveyed by the branching conveyance unit **90** to either the non-inverting unit **91** or the inverting unit **92**.

(93) The gate unit **94** performs sorting by switching the conveyance route so as to convey, among the banknotes S conveyed by the branching conveyance unit **90**, the banknotes S in the first front orientation N1 to the non-inverting unit **91**, and convey the remaining banknotes S in the first back orientation N2 to the inverting unit **92**. The operation of the gate unit **94** is controlled by the control unit **51**. Also, the control unit **51** controls the gate unit **94** based on the identification information.

(94) As a result, among the banknotes S conveyed by the branching conveyance unit the first front-back inverting unit **81** causes the banknotes S in the first front orientation N1 to be discharged from the non-inverting unit **91** while maintaining the first front orientation N1 due to the two front-back inversions by the non-inverting unit **91**. On the other hand, for the banknotes S in the first back orientation N2, the first front-back inverting unit **81** discharges the banknotes S from the inverting unit **92** in the state of the front-back directionality thereof being changed to the first front orientation N1 due to the one front-back inversion by the inverting unit **92**.

(95) Therefore, the first front-back inverting unit **81** can unify the front-back directionalities of the banknotes S discharged from the non-inverting unit **91** and the inverting unit **92** to the first front orientation N1.

(96) (Second Front-Back Inverting Unit)

(97) The second front-back inverting unit **82** is arranged on the left side LH of the first front-back inverting unit **81** and has the same configuration as the first front-back inverting unit **81**.

Accordingly, a detailed description of the second front-back inverting unit **82** will be omitted, with the main points that differ from the first front-back inverting unit **81** being described.

(98) In addition, the same reference numerals are given to the same components of the second inverting unit **82** as the components constituting the first inverting unit **81**.

(99) The branch sorting unit **93** in the second inverting unit **82** sorts, among the banknotes conveyed by the coupled conveyance unit **70**, the banknotes S that have passed through the branch sorting unit **93** in the first inverting unit **81**, that is, the banknotes S in the second front orientation N3 and the banknotes S in the second back orientation N4, so as to flow to the second front-back inverting unit **82** side.

(100) When the banknotes S other than the banknotes S in the second front orientation N3 and the banknotes S in the second back orientation N4 are conveyed by the coupled conveyance unit **70**, the branch sorting unit **93** operates so as to allow these banknotes S to pass to flow toward the left end side, which is the downstream side of the coupled conveyance unit **70**. As a result, even if banknotes S other than the banknotes S in the second front orientation N3 and the banknotes S in the second back orientation N4 have been conveyed to the second front-back inverting unit **82**, these banknotes can be discharged from the option outlet **71** to the outside of the machine or to an evacuation pocket or the like.

(101) The gate unit **94** in the second front-back inverting unit **82** performs sorting by switching the conveyance route so as to convey, among the banknotes S conveyed by the branching conveyance unit **90**, the banknotes S in the second front orientation N3 to the non-inverting unit **91**, and convey the remaining banknotes S in the second back orientation N4 to the inverting unit **92**.

(102) As a result, among the banknotes S conveyed by the branching conveyance unit **90**, the

second front-back inverting unit **82** causes the banknotes S in the second front orientation N3 to be discharged from the non-inverting unit **91** while maintaining the second front orientation N3 due to the two front-back inversions by the non-inverting unit **91**. On the other hand, for the banknotes S in the second back orientation N4, the second front-back inverting unit **82** discharges the banknotes S from the inverting unit **92** in the state of the front-back directionality thereof being changed to the second front orientation N3 due to the one front-back inversion by the inverting unit **92**.

(103) Therefore, the second front-back inverting unit **82** can unify the front-back directionalities of the banknotes S discharged from the non-inverting unit **91** and the inverting unit **92** to the second front orientation N3.

(104) As shown in FIGS. 2 and 7, the accumulation and bundling device **3** of the present embodiment further includes a first accumulation and storage unit **100**, a second accumulation and storage unit **101**, a bundling unit **103**, and an accumulation conveyance unit **104**. The first accumulation and storage unit **100** accumulates and stores the banknotes S inverted by the first front-back inverting unit **81**. The second accumulation and storage unit **101** accumulates and stores the banknotes S inverted by the second front-back inverting unit **82**. The bundling unit **103** binds the accumulated banknotes S into a small bundle. The accumulation conveyance unit **104** conveys the banknotes S stored in the first accumulation and storage unit **100** and the banknotes S stored in the second accumulation and storage unit **101** to the bundling unit **103** while maintaining the accumulation state.

(105) The first accumulation and storage unit **100**, the second accumulation and storage unit **101**, the bundling unit **103**, and the accumulation conveyance unit **104** are provided in the device case **60** so as to be positioned below the first and second inverting units **81** and **82**.

(106) (First Accumulation and Storage Unit)

(107) The first accumulation and storage unit **100** is positioned below the branching conveyance unit **90** in the first inverting unit **81** and between the left end of the third non-inverting unit **91c** and the right end of the third inverting unit **92c**. Thereby, the banknotes S in the first front orientation N1 that are discharged from the third non-inverting unit **91c** after two front-back inversions by the non-inverting unit **91** are loaded one by one into the first accumulation and storage unit **100**.

Moreover, the banknotes S in the first front orientation N1 that are discharged from the third inverting unit **92c** after one front-back inversion by the inverting unit **92** are loaded one by one into the first accumulation and storage unit **100**.

(108) Accordingly, the banknotes S unified in the first front orientation N1 from the non-inverting unit **91** and the inverting unit **92** are continuously and alternately loaded into the first accumulation and storage unit **100**, for example. Therefore, the banknotes S in the first front orientation N1 can be stored in an accumulated state the first accumulation and storage unit **100**, and the banknotes S can be grouped by a predetermined number (for example, 100).

(109) In particular, a predetermined number of the banknotes S can be accumulated in the first accumulation and storage unit **100** in a state of being aligned in the horizontal width direction and the vertical width direction of the banknotes S. At this time, as described above, the banknotes S can be accumulated in a state of the horizontal width direction of the banknotes S being made to match the front-rear direction L3 of the apparatus, and the vertical width direction of the banknotes S being made to match the left-right direction L1 of the apparatus.

(110) The first accumulation and storage unit **100** will be described in detail below.

(111) As shown in FIG. 8, the accumulation and storage unit **100** is provided with a unit base unit (accumulation unit according to the present invention) that accumulates the banknotes S so that a first direction, which is one direction of the vertical width direction and the horizontal width direction of the banknotes S, and a second direction, which is the other direction, face in predetermined directions. In the present embodiment, the unit base unit **110** accumulates the banknotes S so that the horizontal width direction of the banknotes S faces the front-rear direction L3 of the apparatus, and the vertical width direction of the banknotes S faces the left-right direction

L1 of the device (that is, the conveyance direction).

(112) Furthermore, the first accumulation and storage unit **100** is provided with a width-direction alignment unit **120** that aligns the banknotes S accumulated in the unit base unit **110** in the horizontal width direction, and a conveyance direction alignment unit **130** that aligns the banknotes S accumulated in the unit base unit **110** in the conveyance direction.

(113) (First Accumulation and Storage Unit: Unit Base Unit)

(114) As shown in FIGS. **8** and **9**, the unit base unit **110** is provided with a lower base unit **111** and an upper base unit **112** arranged above the lower base unit **111** with a space therebetween. The lower base unit **111** and the upper base unit **112** are fixed with the space therebetween in the vertical direction L2 by a supporting member or the like (not shown).

(115) The lower base unit **111** is made of, for example, a thin metal plate, and has a rectangular outer shape in plan view that is longer in the front-rear direction L3 than in the left-right direction L1. Note that the lower base unit **111** is formed by bending the metal plate.

(116) Like the lower base unit **111**, the upper base unit **112** is made of, for example, a thin metal plate, and has a rectangular outer shape in plan view that is longer in the front-rear direction L3 than in the left-right direction L1. However, the upper base unit **112** is formed to have a smaller outer shape than the lower base unit **111**. Further, the upper base unit **112** is formed by bending a metal plate, and is appropriately formed with hollowed recesses and the like.

(117) As shown in FIG. **8**, in the upper surface of the upper base unit **112**, the central area positioned at the center in the left-right direction L1 is used as a main accumulation area **113** where banknotes S discharged from the non-inverting unit **91** and the inverting unit **92** are accumulated. Therefore, it is possible to accumulate banknotes S on the upper base unit **112** using the main accumulation area **113**.

(118) At both ends of the upper base unit **112** in the left-right direction L1, there are formed a plurality of expansion accommodation holes **114** that pass through the upper base unit **112** in the vertical direction L2 and open on both sides in the left-right direction L1. The plurality of expansion accommodation holes **114** are formed so as to be spaced apart in the front-rear direction L3.

(119) The conveyance direction alignment unit **130** is accommodated in the plurality of expansion accommodation holes **114** so as to be relatively movable in the left-right direction L1 with respect to the upper base unit **112**.

(120) Specifically, a third stopper unit **270**, an alignment plate **311** and the like of a third alignment mechanism **131**, to be described later, are mainly accommodated to be movable in the left-right direction L1 in the plurality of expansion accommodation holes **114** formed in the right side RH of the upper base unit **112**. A fourth stopper unit **420** of a fourth alignment mechanism **132**, which will be described later, is accommodated so as to be movable in the left-right direction L1 in a plurality of expansion accommodation holes (not shown) formed in the left side LH of the upper base unit **112**.

(121) These will be explained later.

(122) The area on the top surface of the upper base **112** that is located on both sides of the main accumulation area **113** in the left-right direction L1 and between adjacent expansion accommodation holes **114** in the front-rear direction L3 is used as an expansion accumulation area **115**.

(123) The expansion accumulation area **115** is used as an additional area in which banknotes S can be accumulated in addition to the main accumulation area **113** when the third alignment mechanism **131** and the fourth alignment mechanism **132** are moved away from each other in the left-right direction L1. Note that the expansion accumulation area **115** can accumulate the banknotes S even if the third alignment mechanism **131** and the fourth alignment mechanism **132** are spaced apart to the maximum extent in the left-right direction L1.

(124) (First Accumulation and Storage Unit: Width-Direction Alignment Unit)

(125) As shown in FIGS. **8** to **10**, the width-direction alignment unit **120** is provided with a first alignment mechanism **121** and a second alignment mechanism **122** arranged to face each other in the front-rear direction **L3** with the upper base unit **112** interposed therebetween, and uses the main accumulation area **113** to align banknotes **S** accumulated on the upper base unit **112** in the horizontal width direction.

(126) The first alignment mechanism **121** is located more to the rear **BK** side of the apparatus than the upper base unit **112** and is provided so as to be movable relative to the upper base unit **112** in the front-rear direction **L3**. In contrast, the second alignment mechanism **122** is located more to the front **FW** side of the apparatus than the upper base unit **112** and is provided so as to be movable relative to the upper base unit **112** in the front-rear direction **L3**.

(127) (Width-Direction Alignment Unit: First Alignment Mechanism)

(128) The first alignment mechanism **121** will be described in detail.

(129) The first alignment mechanism **121** is provided with a first base unit **140**, a first base drive unit **150**, a first alignment unit **160**, and a first alignment drive unit **170**. The first base unit **140** is movable in the front-rear direction **L3** with respect to the upper base unit **112**. The first alignment unit **160** is provided in the first base unit **140**. The first base drive unit **150** drives the first base unit **140**. The first alignment drive unit **170** drives the first alignment unit **160**.

(130) The first base unit **140** is made of, for example, a thin metal plate. The first base unit **140** is arranged between the lower base unit **111** and the upper base unit **112** and is arranged on the rear **BK** side of the upper base unit **112**.

(131) On the underside of the first base unit **140**, a first guide shaft **141** fixed to the lower base unit **111** is inserted inside, and a first linear motion guide unit **142** is provided to linearly guide the first guide shaft **141**.

(132) The first guide shaft **141** is a cylindrical rod arranged on the right side **RH** of the lower base unit **111** and linearly extending along the front-rear direction **L3**. The first linear motion guide unit **142** is a guide member such as a linear ball bearing, for example. The first linear motion guide unit **142** is arranged on the right side **RH** of the first base unit **140**, and the first guide shaft **141** is slidably fitted therein. Thereby, the first base unit **140** is supported so as to be movable along the front-rear direction **L3** with high-precision straightness while being guided by the first guide shaft **141**.

(133) The first base drive unit **150** is provided with a first rack gear **151**, a first pinion gear **152**, and a first base drive motor **153**. The first rack gear **151** is fixed to the bottom surface of the first base unit **140**. The first pinion gear **152** is provided in the first base unit **140** side and meshes with the first rack gear **151**. The first base drive motor **153** drives the first pinion gear **152**.

(134) The first rack gear **151** is formed in a straight line extending along the front-rear direction **L3**. The first rack gear **151** is arranged on the left side **LH** of the first base unit **140** and fixed to the lower surface of the first base unit **140** with the rack teeth directed to the right side **RH**. The first base drive motor **153** is fixed to the lower surface side of the lower base unit **111**, with the drive shaft thereof protruding to the upper surface side of the lower base. The first pinion gear **152** is attached to the drive shaft of the first base drive motor **153**.

(135) The first base drive motor **153** is, for example, a stepping motor, and its operation is controlled by the control unit **51**, and rotates the first pinion gear **152** forward and backward based on signals from the control unit **51**. Thereby, the first base unit **140** can be moved in the front-rear direction **L3** while being guided by the first guide shaft **141** via the first rack gear **151**.

(136) In this way, since the first base unit **140** can be moved in the front-rear direction **L3**, by moving the first base unit **140** according to the size (horizontal width) of the banknotes **S**, banknotes **S** of various sizes can be handled.

(137) The first alignment unit **160** has a first arm unit **161** and a first plane contact unit **162**, and is formed in a Z shape (crank shape) in plan view. The first alignment unit **160** is arranged on the upper surface side of the first base unit **140**. Also, the first alignment unit **160** is arranged so as to

be positioned above the upper base unit **112** of the unit base unit **110**. The first arm unit **161** is attached to a first alignment shaft **163** rotatably erected on the upper surface of the first base unit **140**. Accordingly, the first alignment unit **160** is rotatable around the first alignment shaft **163**. The first alignment shaft **163** is arranged so as to be positioned on the rear BK side of the first rack gear **151**.

(138) The first alignment unit **160** is rotatable (movable) about the first alignment shaft **163** between a first retracted position **P1** as shown in FIG. **10** and a first alignment position **K1** as shown in FIG. **11**. As shown in FIG. **10**, the first retracted position **P1** is a position where a first planar contact unit **162** of the first alignment unit **160** is located more toward the rear BK side than an entry area of the banknotes **S** through which the banknotes **S** discharged from the non-inverting unit **91** and the inverting unit **92** pass until being accumulated on the upper base unit **112**, with the first planar contact unit **162** being separated to the rear BK side from the banknotes **S** accumulated on the upper base unit **112**. As shown in FIG. **11**, the first alignment position **K1** is a position where the first planar contact unit **162** of the first alignment unit **160** contacts one edge in the horizontal width direction of the banknotes **S** accumulated on the upper base unit **112**.

(139) As shown in FIGS. **9** and **10**, the first planar contact unit **162** is provided at the tip of the first arm unit **161**. The first planar contact unit **162** is wider than the first arm unit **161** and formed in a flat planar shape. The first planar contact unit **162** is formed so that the lower end thereof is positioned at the same height as the upper surface of the upper base unit **112** and the upper end is located above the height of all the banknotes **S** when a predetermined number (for example, 100) of banknotes **S** are accumulated on the upper base unit **112**.

(140) In particular, when the first alignment unit **160** is positioned at the first alignment position **K1**, the first planar contact unit **162** is provided so as to be substantially parallel along the vertical width direction (conveyance direction) of the banknotes **S**, as shown in FIG. **11**. Thereby, the first planar contact unit **162** can come into, for example, linear contact or surface contact with one edge in the horizontal width direction of the banknotes **S**.

(141) In addition, since the first base unit **140** can be moved according to the horizontal width size of the banknotes **S** accumulated on the upper base unit **112**, the rotation angle of the first alignment unit **160** around the first alignment shaft **163** can be made constant regardless of the horizontal width size of the banknotes **S** to be accumulated.

(142) As shown in FIGS. **9** and **10**, the first alignment drive unit **170** is provided with a first alignment gear **171**, a first motor gear **172**, and a first alignment drive motor **173**. The first alignment gear **171** is provided integrally with the first alignment shaft **163**. The first motor gear **172** meshes with the first alignment gear **171**. The first alignment drive motor **173** drives the first motor gear **172**.

(143) The first alignment gear **171** is formed in a fan shape in a plan view with the teeth facing the right side **RH**, and is arranged on the upper surface side of the first base unit **140**. The first alignment drive motor **173** is fixed to the lower surface side of the first base unit **140**, with the drive shaft thereof protruding to the upper surface side of the first base unit **140**. The first motor gear **172** is attached to the drive shaft of the first alignment drive motor **173**.

(144) The first alignment drive motor **173** is, for example, a stepping motor. The operation of the first alignment drive motor **173** is controlled by the control unit **51**, with the first alignment gear **171** being driven forward and backward on the basis of a signal from the control unit **51**. Accordingly, the first alignment drive motor **173** can rotate the first alignment shaft **163** via the first alignment gear **171**, and thereby can move the first alignment unit **160** back and forth between the first retracted position **P1** and the first alignment position **K1**.

(145) (Width-Direction Alignment Unit: Second Alignment Mechanism)

(146) The second alignment mechanism **122** is arranged more on the front **FW** side of the apparatus than the upper base unit **112**. The second alignment mechanism **122** is arranged to face the above-described first alignment mechanism **121** in the front-rear direction **L3** with the upper base unit **112**

interposed therebetween. The second alignment mechanism **122** of this embodiment has the same configuration as the first alignment mechanism **121**. The second alignment mechanism **122** is arranged to be point symmetrical to the first alignment mechanism **121**, rotated 180 degrees, with the reference position O (see FIG. **10**) of the upper base unit **112**, that is, the center of the front-rear direction L3 and the left-right direction L1, as a base point.

(147) Accordingly, with respect to the second alignment mechanism **122**, the same components as those of the first alignment mechanism **121** are given “second” instead of “first” in their names, and will be briefly described below.

(148) The second alignment mechanism **122** is provided with a second base unit **180**, a second base drive unit **190**, a second alignment unit **200**, and a second alignment drive unit **210**. The second base unit **180** is movable in the front-rear direction L3 with respect to the upper base unit **112**. The second base drive unit **190** drives the second base unit **180**. The second alignment unit **200** is provided on the second base unit **180**. The second alignment drive unit **210** drives the second alignment unit **200**.

(149) The second base unit **180** is arranged on the front FW side of the upper base unit **112**. A second linear guide unit **182** that guides a second guide shaft **181** fixed to the lower base unit **111** is provided on the lower surface of the second base unit **180**. The second guide shaft **181** is arranged on the left side LH of the lower base unit **111**, and the second linear guide unit **182** is arranged on the left side LH of the second base unit **180**. Thereby, the second base unit **180** is supported so as to be movable along the front-rear direction L3 with high-precision straightness while being guided by the second guide shaft **181**.

(150) The second base drive unit **190** is provided with a second rack gear **191**, a second pinion gear **192**, and a second base drive motor **193**. The second rack gear **191** is fixed to the bottom surface of the second base unit **180**. The second pinion gear **192** is provided in the second base unit **180** and meshes with the second rack gear **191**. The second base drive motor **193** drives the second pinion gear **192**.

(151) The second rack gear **191** is arranged on the right side RH of the second base unit **180**. The second pinion gear **192** is attached to the drive shaft of the second base drive motor **193**. The second base drive motor **193** rotates the second pinion gear **192** forward and backward based on a signal from the control unit **51**. Thereby, the second base unit **180** can be moved in the front-rear direction L3 while being guided by the second guide shaft **181** via the second rack gear **191**.

(152) The second alignment unit **200** has a second arm unit **201** and a second planar contact unit **202** and is arranged on the upper surface side of the second base unit **180**. Also, the second alignment unit **200** is arranged so as to be positioned above the upper base unit **112** of the unit base unit **110**. The second arm unit **201** is attached to a second alignment shaft **203** rotatably erected on the upper surface of the second base unit **180**. Accordingly, the second alignment unit **200** is rotatable around the second alignment shaft **203**. The second alignment shaft **203** is arranged so as to be positioned on the front FW side of the second rack gear **191**.

(153) The second alignment unit **200** is rotatable (movable) about the second alignment shaft **203** between a second retracted position P2 as shown in FIG. **10** and a second alignment position K2 as shown in FIG. **11**. As shown in FIG. **10**, the second retracted position P2 is a position where the second planar contact unit **202** of the second alignment unit **200** is located more toward the front FW side than an entry area of the banknotes S through which the banknotes S discharged from the non-inverting unit **91** and the inverting unit **92** pass until being accumulated on the upper base unit **112**, with the second planar contact unit **202** being separated to the front FW side from the banknotes S accumulated on the upper base unit **112**. As shown in FIG. **11**, the second alignment position K2 is a position where the second planar contact unit **202** of the second alignment unit **200** contacts the other edge in the horizontal width direction of the banknotes S accumulated on the upper base unit **112**.

(154) As shown in FIGS. **9** and **10**, the second planar contact unit **202** is provided at the tip of the

second arm unit **201**. The second planar contact unit **202** is formed so that the lower end thereof is positioned at the same height as the upper surface of the upper base unit **112** and the upper end is located above the height of all the banknotes S when a predetermined number (for example, 100) of banknotes S are accumulated on the upper base unit **112**.

(155) In particular, when the second alignment unit **200** is positioned at the second alignment position **K2**, the second planar contact unit **202** is provided so as to be substantially parallel along the vertical width direction (conveyance direction) of the banknotes S, as shown in FIG. **11**.

Thereby, the second planar contact unit **202** can come into, for example, linear contact or surface contact with the other edge in the horizontal width direction of the banknotes S.

(156) As shown in FIGS. **9** and **10**, the second alignment drive unit **210** is provided with a second alignment gear **211**, a second motor gear **212**, and a second alignment drive motor **213**. The second alignment gear **211** is provided integrally with the second alignment shaft **203**. The second motor gear **212** meshes with the second alignment gear **211**. The second alignment drive motor **213** drives the second motor gear **212**.

(157) The second alignment gear **211** is formed in a fan shape in a plan view with the teeth facing the left side **LH**, and is arranged on the upper surface side of the second base unit **180**. The second motor gear **212** is attached to the drive shaft of the second alignment drive motor **213**.

(158) The second alignment drive motor **213** rotates the second alignment gear **211** forward and backward based on a signal from the control unit **51**. Accordingly, the second alignment shaft **203** can be rotated via the second alignment gear **211**, and the second alignment unit **200** can be made to move back and forth between the second retracted position **P2** and the second alignment position **K2**.

(159) (Control of Width-Direction Alignment Unit)

(160) The control unit **51** controls the operation of the width-direction alignment unit **120** equipped with the first alignment mechanism **121** and the second alignment mechanism **122** configured as described above. Note that the control unit **51** performs appropriate control while detecting banknotes S based on various sensors (for example, non-contact sensors such as photoelectric sensors, contact sensors, and the like) and the operating positions of the first alignment mechanism **121** and second alignment mechanism **122**.

(161) In particular, the control unit **51**, according to the denomination of the banknote S preset by the operation display unit **50**, moves the first base unit **140** and the second base unit **180** along the front-rear direction **L3** to adjust the spacing between the first alignment unit **160** and the second alignment unit **200**. Thereby, according to the width of the banknotes S, the first alignment unit **160** and the second alignment unit **200** can be arranged facing each other in the front-rear direction **L3** at an optimum spacing.

(162) More specifically, the control unit **51** controls the first base unit **140** and the second base unit **180** to move along the front-rear direction **L3** so that the interval in the front-rear direction **L3** between the first planar contact unit **162** and the second planar contact unit **202** matches the horizontal width of the banknotes S to be accumulated on the upper base unit **112** when the first alignment unit **160** and the second alignment unit **200** have been positioned in the first alignment position **K1** and second alignment position **K2**.

(163) As a result, it is possible to flexibly deal with banknotes S having small horizontal widths as shown in FIG. **11** and banknotes S having large horizontal widths as shown in FIG. **12**.

(164) Subsequently, the control unit **51** performs control so as to rotate the first alignment unit **160** and the second alignment unit **200** from the first retracted position **P1** and the second retracted position **P2** to the first alignment position **K1** and the second alignment position **K2**, respectively, to tap the banknotes S accumulated in the upper base unit **112** from both sides in the horizontal width direction, and thereby align the banknotes S in the horizontal width direction.

(165) At this time, in the present embodiment, the control unit **51** performs control so as to simultaneously rotate the first alignment unit **160** and the second alignment unit **200** to so as to

shift from the first retracted position P1 and the second retracted position P2 to the first alignment position K1 and the second alignment position K1.

(166) (First Accumulation and Storage Unit: Conveyance Direction Alignment Unit)

(167) As shown in FIG. 8, the conveyance direction alignment unit 130 is provided with a third alignment mechanism 131 and a fourth alignment mechanism 132 arranged to face each other in the left-right direction L1 with the upper base unit 112 interposed therebetween. The conveyance direction alignment unit 130 mainly uses the main accumulation area 113 to align the banknotes S accumulated on the upper base unit 112 in the conveyance direction (vertical width direction).

(168) The third alignment mechanism 131 is located more on the right side RH of the apparatus than the upper base unit 112 and is provided so as to be movable relative to the upper base unit 112 in the left-right direction L1. In contrast, the fourth alignment mechanism 132 is located more on the left side LH of the apparatus than the upper base unit 112 and is provided so as to be movable relative to the upper base unit 112 in the left-right direction L1.

(169) (Conveyance Direction Alignment Unit: Third Alignment Mechanism)

(170) As shown in FIGS. 8, 13 and 14, the third alignment mechanism 131 is provided with a third base unit 250, a third base drive unit 260, a third stopper unit 270 provided on the third base unit 250, a third conveyance unit (third accumulation conveyance unit according to the present invention) 280 provided in the third base unit 250, and a third alignment unit 310. The third base unit 250 is movable in the left-right direction L1 with respect to the upper base unit 112. The third base drive unit 260 drives the third base unit 250.

(171) The third base unit 250 is, for example, a thin metal plate, and is arranged above the upper base unit 112 and is arranged on the right side RH of the upper base unit 112. The third base unit 250 has a front wall 251 and a rear wall 252 facing each other in the front-rear direction L3, and a right wall 253 coupling the front wall 251 and the rear wall 252 in the front-rear direction L3. The front wall 251 and the rear wall 252 are arranged on the outer side of the upper base unit 112 in the front-rear direction L3. Furthermore, the right wall 253 is arranged on the right side RH of the upper base unit 112.

(172) In the front wall 251 of the third base unit 250 is provided a third linear motion guide unit 256 through which a guide shaft 255 that is fixed in the device case 60 is inserted to linearly guide the guide shaft 255.

(173) The guide shaft 255 is a cylindrical rod arranged more to the front FW side than the front wall 251 and linearly extending along the left-right direction L1. The third linear motion guide unit 256 is a guide member such as a linear ball bearing, which is arranged in the front wall 251 and has the guide shaft 255 slidably fitted therein. Thereby, the third base unit 250 is supported so as to be movable along the left-right direction L1 with high-precision straightness while being guided by the guide shaft 255. A plurality of stopper mounting units 257 projecting toward the left side LH are formed in the right wall 253 of the third base unit 250. The plurality of stopper mounting units 257 are arranged at spacings in the front-rear direction L3 corresponding to the plurality of expansion accommodation holes 114 formed in the right end portion of the upper base unit 112.

(174) The plurality of stopper mounting units 257 enter into each of the plurality of expansion accommodation holes 114 from the right side RH. A third stopper unit 270 is fixed to each of the plurality of stopper mounting units 257. The third stopper unit 270 is formed vertically extending in the vertical direction L2, receives the banknotes S discharged from the inverting unit 92 side, and guides the banknotes S toward the upper base unit 112.

(175) In the illustrated example, the case where the third base unit 250 is provided with four third stopper units 270 is taken as an example. These third stopper units 270 are movable within the expansion accommodation hole 114 as the third base unit 250 moves in the left-right direction L1.

(176) The third base drive unit 260 is provided with a third rack gear 261, a third pinion gear 262, and a third base drive motor 263. The third rack gear 261 is fixed to the front wall 251 of the third base unit 250. The third pinion gear 262 meshes with the third rack gear 261. The third base drive

motor **263** drives the third pinion gear **262** directly or indirectly via a base relay gear or the like. (177) The third rack gear **261** is formed in a linear shape extending along the left-right direction **L1**, and is fixed to the right wall **253** with the rack teeth directed downward. The third base drive motor **263** is, for example, a stepping motor, and its operation is controlled by the control unit **51**, and rotates the third pinion gear **262** forward and backward based on signals from the control unit **51**. As a result, the third base unit **250** can be moved in the left-right direction **L1** while being guided by the guide shaft **255** via the third rack gear **261**.

(178) In this manner, the third base unit **250** can be moved in the left-right direction **L1**. By moving the third base unit **250** according to the size (vertical width) of the banknote **S**, it is possible to handle banknotes **S** of various sizes.

(179) The third conveyance unit **280** is provided at the upper unit of the third base unit **250**, and in addition to receiving the banknotes **S** discharged from the non-inverting unit **91** also loads the banknotes **S** into the upper base unit **112**. The third conveyance unit **280** is provided with a third conveyance path **281**, a third conveyance roller **282**, a third impeller **283**, and a third conveyance drive unit **285**.

(180) As shown in FIG. **14**, the third conveyance path **281** has a third upper conveyance guide **281a** and a third lower conveyance guide **281b** arranged with a gap in the vertical direction **L2**.

(181) As shown in FIGS. **8** and **13**, the third lower conveyance guide **281b** is arranged on the upper side of the third base unit **250**. The third lower conveyance guide **281b** is fixed so as to span between the front wall **251** and the rear wall **252**. The third lower conveyance guide **281b** is formed in a rectangular shape in plan view that is longer in the front-rear direction **L3** than in the left-right direction **L1**, with the upper surface serving as a guide surface for guiding the banknotes **S**.

(182) A plurality of slit holes **290** are formed in the third lower conveyance guide **281b** so as to pass through the third lower conveyance guide **281b** in the vertical direction **L2** and open to the left side **LH**. The slit holes **290** are formed in a horizontally long shape extending along the left-right direction **L1**, and are formed in plurality at intervals in the front-rear direction **L3**.

(183) Joint projections **291** are formed at the right end of the third lower conveyance guide **281b** so as to extend toward the right side **RH**. The joint projections **291** are formed to extend linearly, and are formed in a plurality at intervals in the front-rear direction **L3**. As a result, the plurality of joint projections **291** are formed in a comb shape. Also, a gap between the joint projections **291** adjacent in the front-rear direction **L3** is a joint accommodation unit **292** that opens to the right side **RH**.

(184) Here, as shown in FIG. **8**, a joint projection **91d** and a joint accommodation unit **91e** are formed in the left end (outlet) of the third non-inverting unit **91c**, which constitutes the non-inverting unit **91**. The joint projection **91d** and the joint accommodation unit **91e** correspond to the joint projection **291** and the joint accommodation unit **292**, respectively, and are combined with the joint projection **291** and the joint accommodation unit **292**, respectively.

(185) The joint projections **291** on the side of the third non-inverting unit **91c** are formed to extend linearly toward the left side **LH**. A plurality of the joint projections **91d** are formed in a comb-teeth shape so as to enter the joint accommodation units **292** on the side of the third lower conveyance guide **281b**. Therefore, the joint projections **91d** on the side of the third non-inverting unit **91c** are accommodated in the joint accommodation units **292** on the side of the third lower conveyance guide **281b** so as to be movable in the left-right direction **L1**.

(186) The joint accommodation units **91e** on the side of the third non-inverting unit **91c** are formed between the joint projections **91d** adjacent to each other in the front-rear direction **L3**, and the joint projections **291** on the side of the third lower conveyance guide **281b** are capable of entering therein. Accordingly, the joint projections **291** on the side of the third lower conveyance guide **281b** are accommodated in the joint accommodation units **91e** on the side of the third non-inverting unit **91c** so as to be movable in the left-right direction **L1**.

(187) As described above, the joint projections **291** and joint accommodation units **292** on the side of the third lower conveyance guide **281b** are combined with the joint projections **91d** and joint

accommodation units **91e** on the side of the third non-inverting unit **91c**. As a result, even if the third alignment mechanism **131** moves in the left-right direction **L1**, which is the conveyance direction, the coupled state between the third lower conveyance guide **281b** and the third non-inverting unit **91c** can be stably maintained without being affected by this movement. Therefore, it is possible to appropriately transfer the banknotes **S** from the third non-inverting unit **91c** to the upper surface of the third lower conveyance guide **281b**.

(188) As shown in FIG. **14**, the third upper conveyance guide **281a** is formed in a rectangular shape in plan view that is longer in the front-rear direction **L3** than in the left-right direction **L1**. The third upper conveyance guide **281a** is arranged to cover the entire third lower conveyance guide **281b** from above, and is fixed so as to span between the front wall **251** and the rear wall **252** of the third base unit **250**. The lower surface of the third upper conveyance guide **281a** serves as a guide surface for guiding the banknotes **S**. Thereby, the banknotes **S** conveyed from the non-inverting unit **91** can be conveyed toward the upper base unit **112** between the upper surface of the third lower conveyance guide **281b** and the lower surface of the third upper conveyance guide **281a**.

(189) A plurality of joint projections and a plurality of joint accommodation units are also formed in the right end unit of the third upper conveyance guide **281a** similarly to the third lower conveyance guide **281b**, and these are combined with the third non-inverting unit **91c**.

(190) Furthermore, the third upper conveyance guide **281a** protrudes toward the left side **LH** more than the third lower conveyance guide **281b**. The third upper conveyance guide **281a** is inclined obliquely downward so as to extend downward toward the upper base unit **112** side. This allows the banknotes **S** conveyed from the non-inverting unit **91** to be reliably guided while the banknotes **S** are conveyed and accumulated as they are dropped toward the upper base unit **112**.

(191) As shown in FIGS. **8** and **13**, the third conveyance roller **282** and the third impeller **283** are each arranged in a slit hole **290** formed in the third lower conveyance guide **281b**. In the illustrated example, two third conveyance rollers **282** and five third impellers **283** are provided. However, the numbers of the third conveyance rollers **282** and the third impellers **283** are not limited to this case, and may be changed as appropriate.

(192) The third conveyance roller **282** and the third impeller **283** are attached to a third conveyance shaft **284** arranged between the front wall **251** and the rear wall **252** of the third base unit **250** along the front-rear direction **L3**. The third conveyance shaft **284** is arranged substantially directly below the third lower conveyance guide **281b** and is rotatably supported by the front wall **251** and the rear wall **252**. Thereby, by rotating the third conveyance shaft **284**, the third conveyance rollers **282** and the third impellers **283** can be rotated, and the banknotes **S** can be conveyed.

(193) The third conveyance drive unit **285** is provided with a third conveyance gear **300** provided on the third conveyance shaft **284**, a third intermediate gear **301** that meshes with the third conveyance gear **300**, and a third conveyance drive motor **302** that drives the third intermediate gear **301**.

(194) The third conveyance shaft **284** protrudes rearward from the rear wall **252** of the third base unit **250** toward the rear **BK** side, and the third conveyance gear **300** is fixed to the rear end unit of the third conveyance shaft **284**. The third relay gear **301** is arranged further to the rear **BK** side than the rear wall **252** of the third base unit **250** and is fixed to the drive shaft of the third conveyance drive motor **302**.

(195) The third conveyance drive motor **302** is, for example, a stepping motor. The operation of the third conveyance drive motor **302** is controlled by the control unit **51**. The third conveyance drive motor **302** rotates the third conveyance shaft **284** via the third relay gear **301** and the third conveyance gear **300** based on a signal from the control unit **51**. Thereby, the third conveyance roller **282** and the third impeller **283** can be rotated together with the third conveyance shaft **284**, and it becomes possible to convey the banknotes **S** that have been conveyed between the third upper conveyance guide **281a** and the third lower conveyance guide **281b** from the non-inverting unit **91** toward the upper base unit **112**.

(196) The third alignment unit **310** is provided with an alignment plate **311** provided in the third base unit **250** and an alignment plate drive unit **330** that drives the alignment plate **311** in conjunction with the third conveyance rollers **282**.

(197) As shown in FIG. **13**, the alignment plate **311** is formed in a vertically long shape extending in the vertical direction **L2**, and is arranged so as to be aligned with the third stopper unit **270** in the left-right direction **L1**. In the illustrated example, two of the alignment plates **311** are arranged adjacent to the two third stopper units **270**. However, the number of alignment plates **311** is not limited to two, and at least one should be arranged.

(198) The alignment plate **311** includes a vertically long contact unit **312** and an alignment plate main body **313**. The contact unit **312** has a flat contact surface facing the upper base unit **112** side. The alignment plate main body **313** is integrally formed with the contact unit **312** and extends more toward the right side **RH** and lower than the third stopper unit **270**. The contact unit **312** can come into contact with one edge positioned in the vertical width direction of the banknotes **S**.

(199) As shown in FIGS. **13** and **15**, a receiving hole **314** penetrating the alignment plate body **313** in the front-rear direction **L3** is formed in the alignment plate body **313**. The receiving hole **314** is formed in a vertically long shape that is longer in the vertical direction **L2** than in the left-right direction **L1**.

(200) The lower end unit of the alignment plate body **313** is rotatably attached to an alignment shaft **320** pivotally supported by the third base unit **250**. The alignment shaft **320** is arranged along the front-rear direction **L3**, and both sides of the alignment shaft **320** are pivotally supported by support pieces **321** integrally formed with the rear wall **252** of the third base unit **250**.

(201) Accordingly, the alignment plate **311** can swing about the alignment shaft **320** in the left-right direction **L1**.

(202) Specifically, the alignment plate **311** is swingable (movable) between a third alignment position **K3** shown in FIGS. **13** and **15** and a third retracted position **P3** shown in FIG. **16**. At the third alignment position **K3**, the contact unit **312** protrudes more to the left side **LH** than the stopper portion and contacts one edge in the vertical width direction of the banknotes **S** accumulated on the upper base unit **112**. At the third retracted position **P3**, the contact unit **312** retracts more to the right side **RH** than the stopper portion and separates from the banknotes **S** accumulated on the upper base unit **112**.

(203) The alignment plate **311** is accommodated together with the third stopper unit **270** in the expansion accommodation hole **114** provided in the upper base unit **112**.

(204) As shown in FIGS. **13** and **15**, the alignment plate drive unit **330** is provided with an alignment plate drive shaft **331**, an eccentric cam **340**, and an alignment plate drive transmission unit **350**.

(205) The alignment plate drive shaft **331** is positioned below the third conveyance shaft **284** and above the alignment shaft **320**, and is arranged between the front wall **251** and the rear wall **252** of the third base unit **250** along the front-rear direction **L3**. The alignment plate drive shaft **331** is rotatably supported by the front wall **251** and the rear wall **252** while passing through the receiving hole **314** formed in the alignment plate **311**. The alignment plate drive shaft **331** projects rearward from the rear wall **252** of the third base unit **250** toward the rear **BK** side.

(206) The eccentric cam **340** is arranged in the vertically elongated receiving hole **314** formed in the alignment plate **311**. The eccentric cam **340** is an eccentric disc cam having an outer diameter that contacts the inner surface of receiving hole **314**. The eccentric cam **340** is formed with an eccentric hole **341** passing through the eccentric cam **340** in the front-rear direction **L3**. The eccentric hole **341** is formed at a position eccentric from the center of the eccentric cam **340**. Thereby, the eccentric cam **340** is formed so that the wall thickness (thickness in the radial direction) varies in the circumferential direction.

(207) The eccentric cam **340** has the alignment plate drive shaft **331** inserted into the eccentric hole **341** and is thereby integrally combined with the alignment plate drive shaft **331**. Therefore, the

eccentric cam **340** eccentrically rotates around the alignment plate drive shaft **331** as the alignment plate drive shaft **331** rotates. Accordingly, the eccentric rotation of the eccentric cam **340** can be used to swing the alignment plate **311** between the third alignment position **K3** and the third retracted position **P3**.

(208) Specifically, the eccentric rotation of the eccentric cam **340** accompanying the rotation of the alignment plate drive shaft **331** causes the maximum wall thickness unit **342** to contact the portion of the inner surface of the receiving hole **314** that is on the contact unit **312** side, as shown in FIG. **15**. Accordingly, the eccentric cam **34** can position the alignment plate **311** at the third alignment position **K3**. In contrast, as shown in FIG. **16**, the eccentric rotation of the eccentric cam **340** accompanying the rotation of the alignment plate drive shaft **331** causes the maximum wall thickness unit **342** to contact the portion of the inner surface of the receiving hole **314** that is on the opposite side of the contact unit **312**. Accordingly, the eccentric cam **34** can position the alignment plate **311** at the third retracted position **P3**.

(209) As shown in FIG. **13**, the alignment plate drive transmission unit **350** has a conveyance branch gear **351**, an alignment plate gear **352**, and an alignment plate relay gear **353**.

(210) The conveyance branch gear **351** is provided on the third conveyance shaft **284** and arranged between the third conveyance gear **300** and the rear wall **252** of the third base unit **250**. Thereby, the conveyance branch gear **351** rotates together with the third conveyance shaft **284**, the third conveyance rollers **282** and the third impellers **283**.

(211) The alignment plate gear **352** is provided at the rear end of the third base unit **250**, protruding more toward the rear **BK** side than the rear wall **252**.

(212) The alignment plate relay gear **353** is arranged between the third conveyance gear **300** and the conveyance branch gear **351** and is pivotally supported by the rear wall **252** of the third base unit **250**. The alignment plate relay gear **353** meshes with each of the third conveyance gear **300** and the conveyance branch gear **351**.

(213) As a result, the alignment plate relay gear **353** can transmit the rotational force of the conveyance branch gear **351** to the alignment plate gear **352**, interlocking with the rotation of the third conveyance shaft **284**, which can rotate the alignment plate drive shaft **331** and eccentric cam **340** in conjunction with the rotation of the third conveyance shaft **284**, the third conveyance rollers **282** and the third impellers **283**. Therefore, the alignment plate **311** can be continuously swung between the third alignment position **K3** and the third retracted position **P3** in conjunction with the rotation of the third conveyance rollers **282** and the like.

(214) (Conveyance Direction Alignment Unit: Fourth Alignment Mechanism)

(215) As shown in FIG. **8**, the fourth alignment mechanism **132** is arranged more on the left side **LH** of the apparatus than the upper base unit **112**, and is positioned across the upper base unit **112** so as to face the third alignment mechanism **131** described above in the left-right direction **L1**. The fourth alignment mechanism **132** of this embodiment basically has the same configuration as the third alignment mechanism **131**. The fourth alignment mechanism **132** is arranged in line-symmetry (left-right symmetry) with respect to the third alignment mechanism **131** when viewed from above.

(216) Accordingly, in the fourth alignment mechanism **132**, the same components as those of the third alignment mechanism **131** will be described simply by adding “fourth” instead of “third” to their names.

(217) However, the fourth alignment mechanism **132** does not have a component corresponding to the third alignment unit **310**.

(218) As shown in FIGS. **8** and **14**, the fourth alignment mechanism **132** is provided with a fourth base unit **400**, a fourth base drive unit **410**, a fourth stopper unit **420** provided in the fourth base unit **400**, and a fourth conveyance unit (fourth accumulation conveyance unit according to the present invention) **430** provided in the fourth base unit **400**. The fourth base unit **400** is movable in the left-right direction **L1** with respect to the upper base unit **112**. The fourth base drive unit **410**

drives the fourth base unit **400**.

(219) The fourth base unit **400** has a front wall **401**, a rear wall **402**, and a left wall **403** coupling the front wall **401** and the rear wall **402** in the front-rear direction **L3**. The left wall **403** has a banknote discharge port **404** that penetrates the left wall **403** in the left-right direction **L1** for discharging banknotes accumulated on the upper surface of the upper base unit **112** to the outside (left side **LH**) while maintaining the accumulated state.

(220) In the front wall **401** of the fourth base unit **400** is provided a fourth linear motion guide unit **406** through which a guide shaft **255** that is fixed in the device case **60** is inserted to linearly guide the guide shaft **255**. Thereby, the fourth base unit **400** is supported so as to be movable along the left-right direction **L1** with high-precision straightness while being guided by the guide shaft **255**.

(221) The guide shaft **255** is a common guide shaft with the third base unit **250**. However, the present invention is not limited to this case, and separate guide shafts may be used to guide the third base unit **250** and the fourth base unit **400**, respectively.

(222) The fourth base drive unit **410** is provided with a fourth rack gear **411**, a fourth pinion gear **412**, and a fourth base drive motor **413**. The fourth rack gear **411** is fixed to the front wall **401** of the fourth base unit **400**. The fourth pinion gear **412** meshes with the fourth rack gear **411**. The fourth base drive motor **413** drives the fourth pinion gear **412** directly or indirectly via a base relay gear or the like.

(223) The fourth base drive motor **413** is, for example, a stepping motor. The operation of the fourth base drive motor **413** is controlled by the control unit **51**, and the fourth base drive motor **413** rotates the fourth pinion gear **412** forward and backward based on a signal from the control unit **51**. As a result, the fourth base unit **400** can be moved in the left-right direction **L1** while being guided by the guide shaft **255** via the fourth rack gear **411**.

(224) In this manner, the fourth base unit **400** can be moved in the left-right direction **L1**. By moving the fourth base unit **400** according to the size (vertical width) of the banknote **S**, it is possible to handle banknotes **S** of various sizes.

(225) The fourth conveyance unit **430** is provided at the upper unit of the fourth base unit **400**, and in addition to receiving the banknotes **S** discharged from the inverting unit **92** also loads the banknotes **S** into the upper base unit **112**. The fourth conveyance unit **430** is provided with a fourth conveyance path **431**, a fourth conveyance roller **432**, a fourth impeller **433**, and a fourth conveyance drive unit **435**.

(226) As shown in FIG. **14**, the fourth conveyance path **431** has a fourth upper conveyance guide **431a** and a fourth lower conveyance guide **431b** arranged with a gap in the vertical direction **L2**.

(227) As shown in FIG. **8**, the fourth lower conveyance guide **431b** is formed with a plurality of slit holes **440** that pass through the fourth lower conveyance guide **431b** in the vertical direction **L2** and open to the right side **RH**. A plurality of joint projections **441** are formed in comb-teeth shape toward the left side **LH** at the left end portion of the fourth lower conveyance guide **431b**. A gap between the joint projections **441** adjacent in the front-rear direction **L3** is a joint accommodation unit **442** that opens to the left side **LH**.

(228) Here, a joint projection **92d** and a joint accommodation unit **92e** are formed in the right end (outlet) of the third inverting unit **92c**, which constitutes the inverting unit **92**. The joint projection **92d** and the joint accommodation unit **92e** correspond to the joint projection **441** and the joint accommodation unit **442**, respectively, and are combined with the joint projection **441** and the joint accommodation unit **442**, respectively. Accordingly, even if the fourth alignment mechanism **132** moves in the left-right direction **L1**, which is the conveyance direction, the coupled state between the fourth lower conveyance guide **431b** and the third inverting unit **92c** can be stably maintained without being affected by this movement, and it is possible to properly pass the banknotes **S** from the third inverting unit **92c** to the upper surface of the fourth lower conveyance guide **431b**.

(229) The fourth conveyance roller **432** and the fourth impeller **433** are arranged in the slit holes **440** formed in the fourth lower conveyance guide **431b**. The fourth impeller **433** is attached to a

fourth conveyance shaft **434** arranged between the front wall **401** and the rear wall **402** of the fourth base unit **400**. Thereby, by rotating the fourth conveyance shaft **434**, the fourth conveyance rollers **432** and the fourth impellers **433** can be rotated, and the banknotes S can be conveyed. (230) The fourth conveyance drive unit **435** is provided with a fourth conveyance gear **450** provided on the fourth conveyance shaft **434**, a fourth intermediate gear **451** that meshes with the fourth conveyance gear **450**, and a fourth conveyance drive motor not shown that drives the fourth intermediate gear **451**.

(231) The fourth conveyance drive motor rotates the fourth conveyance shaft **434** via the fourth relay gear **451** and the fourth conveyance gear **450** based on a signal from the control unit **51**. Thereby, the fourth conveyance roller **432** and the fourth impeller **433** can be rotated together with the fourth conveyance shaft **434**, and it becomes possible to convey the banknotes S that have been conveyed between the fourth upper conveyance guide **431a** and the fourth lower conveyance guide **431b** from the inverting unit **92** toward the upper base unit **112**.

(232) By the way, between the front wall **401** and the rear wall **402** of the fourth base unit **400**, a stopper shaft **460** extending along the front-rear direction L3 is supported. The stopper shaft **460** is arranged below the fourth conveyance shaft **434**.

(233) A plurality of fourth stopper units **420** are fixed to the stopper shaft **460** via attachment units **461** attached to the stopper shaft **460**. The plurality of fourth stopper units **420** are arranged at intervals in the front-rear direction L3, and enter the plurality of expansion accommodation holes **114** formed in the upper base unit **112** from the left side LH. These fourth stopper units **420** are formed vertically extending in the vertical direction L2, and play the role of receiving the banknotes S discharged from the non-inverting unit **91** side, and guiding the banknotes S toward the upper base unit **112**.

(234) It should be noted that four fourth stopper units **420** are provided, for example, similarly to the third stopper unit **270**. However, the number of fourth stopper units **420** is not limited to this case, and may be changed as appropriate.

(235) The stopper shaft **460** to which the fourth stopper units **420** are fixed is rotationally driven by a fourth stopper drive unit (not shown). The fourth stopper drive unit has a stopper gear provided on the stopper shaft **460** and a stopper drive motor that drives the stopper gear directly or indirectly via a stopper intermediate gear or the like.

(236) The stopper drive motor is, for example, a stepping motor. The operation of the stopper drive motor is controlled by the control unit **51**, and the stopper drive motor rotates the stopper gear forward and backward based on a signal from the control unit **51**. Accordingly, by rotating the stopper shaft **460**, it is possible to rotate the fourth stopper units **420** around the stopper shaft **460**.

(237) Specifically, the fourth stopper unit **420** can be rotated between the closed position P4 (see FIG. 14) and the open position. At the closed position P4, the fourth stopper unit **420** closes the banknote discharge port **404** and receives the banknotes S discharged from the non-inverting unit **91** side. At the open position, the fourth stopper unit **420** can open the banknote discharge port **404** and discharge the banknotes S accumulated on the upper base unit **112** to the outside.

(238) Note that when positioned at the closed position P4, the fourth stopper unit **420** receives the banknotes S discharged from the non-inverting unit **91** side and guides the banknotes S to the upper base unit **112**.

(239) (Control of Conveyance Direction Alignment Unit)

(240) The control unit **51** controls the operation of the conveyance direction alignment unit **130** equipped with the third alignment mechanism **131** and the fourth alignment mechanism **132** configured as described above. Note that the control unit **51** performs appropriate control while detecting banknotes S based on various sensors (for example, non-contact sensors such as photoelectric sensors, contact sensors, and the like) and the operating positions of the third alignment mechanism **131** and fourth alignment mechanism **132**.

(241) In particular, the control unit **51**, according to the denomination of the banknote S preset by

the operation display unit **50**, moves the third base unit **250** and the third base unit **250** along the left-right direction **L1**, to adjust the spacing between the third stopper unit **270** and the fourth stopper unit **420**. Thereby, according to the vertical width of the banknote **S**, the third stopper unit **270** and the fourth stopper unit **420** can be positioned facing each other in the left-right direction **L1** by the optimal spacing.

(242) As a result, it is possible to flexibly deal with banknotes **S** having small vertical widths as shown in FIG. **14** and banknotes **S** having large vertical widths as shown in FIG. **17**. In particular, in the case shown in FIG. **17**, banknotes **S** can be accumulated using the expansion accumulation area **115** in addition to the main accumulation area **113**.

(243) Moreover, in the state of the fourth stopper units **420** being positioned at the closed unit **P4**, the control unit **51**, by causing the alignment plate **311** to swing from the third retracted position **P3** to the third alignment position **K3**, taps banknotes **S** accumulated on the upper base unit **112** from one side in the vertical width direction. By this operation, the banknotes **S** are aligned in the vertical width direction between the fourth stopper units **420**.

(244) (Second Accumulation and Storage Unit)

(245) As shown in FIG. **2**, the second accumulation and storage unit **101** is arranged on the left side **LH** of the first accumulation and storage unit **100** with the accumulation conveyance unit **104** interposed therebetween, and has the same configuration as the first accumulation and storage unit **100**. Therefore, a detailed description of the second accumulation and storage unit **101** is omitted.

(246) However, since the second accumulation and storage unit **101** is arranged on the left side **LH** of the accumulation conveyance unit **104**, the banknote discharge port **404** opens to the right side **RH**.

(247) (Accumulation Conveyance Unit)

(248) As shown in FIG. **2**, the accumulation conveyance unit **104** is arranged between the first accumulation and storage unit **100** and the second accumulation and storage unit **101** configured as described above. Accordingly, the first accumulation and storage unit **100**, the second accumulation and storage unit **101**, and the accumulation conveyance unit **104** are arranged side by side in the left-right direction **L1**.

(249) The accumulation conveyance unit **104** has, for example, a first chuck member **500** capable of clamping the accumulated banknotes **S** in the vertical direction **L2**. The first chuck member **500** is provided with a first chuck unit **501** located above the accumulated banknotes **S** and a second chuck unit **502** located below the accumulated banknotes **S**. The first chuck unit **501** and the second chuck unit **502** can be relatively moved closer to and away from each other in the vertical direction **L2**, and can hold the accumulated banknotes **S** in the vertical direction **L2** while maintaining the orientation thereof.

(250) The first chuck member **500** configured in this manner is movable between the first accumulation and storage unit **100** and the second accumulation and storage unit **101** along the left-right direction **L1**. Therefore, the accumulated banknotes **S** stored in the first accumulation and storage unit **100** can be taken out through the banknote discharge port **404**, and the accumulated banknotes **S** stored in the second accumulation and storage unit **101** can be taken out through the banknote discharge port **404**.

(251) Further, the first chuck member **500** can transfer the accumulated banknotes **S** taken out from the first accumulation and storage unit **100** and the second accumulation and storage unit **101** to the second chuck member **510** of the bundling unit **103**.

(252) (Bundling Unit)

(253) The bundling unit **103** is arranged below the accumulation conveyance unit **104** and includes the second chuck member **510** capable of receiving banknotes **S** in an accumulated state from the first chuck member **500** of the accumulation conveyance unit **104**.

(254) The second chuck member **510** is provided with a first chuck unit **511** located above the accumulated banknotes **S** and a second chuck unit **512** located below the accumulated banknotes **S**.

The first chuck unit **511** and the second chuck unit **512** can be relatively moved closer to and away from each other in the vertical direction **L2**, and can hold the accumulated banknotes **S** in the vertical direction **L2** while maintaining the orientation thereof.

(255) The bundling unit **103** is further provided with a bundling mechanism unit **520** that bundles the banknotes **S** in the accumulated state sandwiched by the second chuck member **510** with a bundling tape (not shown), thereby creating a small bundle in which a plurality of banknotes **S** are bundled into a single bundle.

(256) The bundling unit **103** further has, for example, a drawer-type small bundle discharge unit **521** for receiving the created small bundle. The small bundle discharge unit **521** is formed on the front surface of the device case **60** as shown in FIG. **1**, and can be pulled out by the operator. Accordingly, the small bundle created by the bundling unit **103** can be taken out through the small bundle discharge unit **521**.

(257) However, the small bundle discharge unit **521** is not limited to a drawer type, and may be provided as an opening. Thus, the small bundle created by the bundling unit **103** can be discharged out of the apparatus through the small bundle discharge unit **521**.

(258) [Action of Banknote Handling Apparatus]

(259) Next, a description will be given of a case where the banknote handling apparatus **1** configured as described above is used to process the banknotes **S** to produce a small bundle of banknotes **S**, for example, 10,000 yen notes.

(260) First, as an initial setting, the operator inputs the denomination of the small bundle to be produced, the number of banknotes **S** to be made into the small bundle, and the like to the operation display unit **50**, and also inputs the number of small bundles to be produced.

(261) After these initial settings are made, the operator loads 10,000 yen banknotes **S** in an accumulated state into the loading unit **11** of the identification and counting device **2**, as shown in FIGS. **1** and **2**. At the time of loading, with regard to the front-back directionalities of the banknotes **S**, the four front-back patterns (first front orientation **N1**, first back orientation **N2**, second front orientation **N3**, and second back orientation **N4**) are mixed.

(262) When the banknotes **S** are set in the loading unit **11**, the banknotes **S** are taken in one by one by the feed-out roller **14** and the take-in roller **15** and transferred to the identification and conveyance unit **20**, as shown in FIG. **2**. Thereby, the banknotes **S** can be conveyed by the first identification and conveyance unit **21** and the second identification and conveyance unit **22**.

(263) During this time, the identification unit **30** counts the conveyed banknotes **S**, identifies the denomination of the banknotes **S**, identifies front and back patterns, and the like, and outputs identification information of the banknotes **S** to the control unit **51**.

(264) Based on the identification information output from the identification unit **30**, the control unit **51**, upon determining that the conveyed banknote **S** can be accepted and that the banknote **S** is of a preset denomination, conveys the banknote **S** from the second identification and conveyance unit **22** to the accumulation and bundling device **3** via the outbound conveyance unit **40**.

(265) Further, the control unit **51**, upon determining that the conveyed banknote **S** is unacceptable based on the identification information output from the identification unit determines the banknote **S** to be a banknote **S** to be rejected, and has the banknote **S** conveyed from the second identification and conveyance unit **22** to the rejection unit **12** via the rejection conveyance unit **41**.

(266) Moreover, the control unit **51**, upon determining that the conveyed banknote **S** can be accepted based on the identification information output from the identification unit **30** and determining that the banknote **S** is of a denomination other than the preset denomination, for example, upon determining that the banknote **S** is a 1,000-yen note different from the 10,000-yen note, determines the banknote **S** to be a banknote **S** to be rejected, and has the banknote **S** conveyed from the second identification and conveyance unit **22** to the rejection unit **12** via the rejection conveyance unit **41**.

(267) As a result, only the 10,000-yen bills **S** determined to be acceptable can be conveyed from

the second identification and conveyance unit **22** to the outbound conveyance unit **40**, while the other banknotes **S** can be conveyed to the rejection unit **12** via the rejection conveyance unit **41**.

(268) As described above, it is possible to use the identification and counting device **2** to extract only banknotes **S** of 10,000 yen notes that are determined to be acceptable and set in advance, and to deliver them to the coupled conveyance unit **70** of the accumulation and bundling device **3**. At this time, the control unit **51** ascertains that each banknote **S** delivered from the identification and counting device **2** to the coupled conveyance unit **70** has one of the four front-rear patterns.

(269) As shown in FIG. 7, when receiving the banknotes **S** from the identification and counting device **2**, the coupled conveyance unit **70** conveys the banknotes **S** toward the downstream side. At this time, the branch sorting unit **93** in the first front-back inverting unit **81** receives an instruction from the control unit **51**, and when the banknote **S** conveyed by the coupled conveyance unit **70** has the front-rear pattern of the first front orientation **N1** and the first back orientation **N2**, sorts the banknote **S** so as to be conveyed to the branching conveyance unit **90**. When the banknote **S** has the front-rear pattern of the second front orientation **N3** and the second back orientation **N4**, the branch sorting unit **93** sorts the banknote **S** toward the second front-back inverting unit **82** side.

(270) The branch sorting unit **93** in the second front-back inverting unit **82** receives an instruction from the control unit **51**, and when the banknote **S** conveyed by the coupled conveyance unit **70** has the front-rear pattern of the second front orientation **N3** and the second back orientation **N4**, sorts the banknote **S** so as to be conveyed to the branching conveyance unit **90**.

(271) When the banknote **S** is conveyed to the branching conveyance unit **90** in the first front-back inverting unit **81**, based on the instruction from the control unit **51**, the gate unit **94** switches the conveyance path so as to convey the banknote **S** in the first front orientation **N1** to the non-inverting unit **91**, and convey the banknote **S** in the first back orientation **N2** to the inverting unit **92**.

(272) As a result, the paper sheet handling apparatus **1** can discharge the banknotes **S** in the first front orientation **N1** from the non-inverting unit **91** while maintaining the first front orientation **N1** as is by inverting the banknotes **S** twice with the non-inverting unit **91**. The paper sheet handling apparatus **1** can discharge the banknotes **S** in the first back orientation **N2** from the inverting unit **92** in a state in which the front-back directionalities have been changed to the first front orientation **N1** by one inversion with the inverting unit **92**.

(273) As a result, the front-back directionalities of the banknotes **S** discharged from the non-inverting unit **91** and the inverting unit **92** can be unified to the first front orientation **N1** by using the first front-back inverting unit **81**. Accordingly, the banknotes **S** whose front-back directionality has been unified to the first front orientation **N1** can be transferred to the first accumulation and storage unit **100**, for example.

(274) When the banknote **S** is conveyed to the branching conveyance unit **90** in the second front-back inverting unit **82**, based on the instruction from the control unit **51**, the gate unit **94** switches the conveyance path so as to convey the banknote **S** in the second front orientation **N3** to the non-inverting unit **91**, and convey the banknote **S** in the second back orientation **N4** to the inverting unit **92**. As a result, the banknote **S** in the second front orientation **N3** can be discharged from the non-inverting unit **91** while maintaining the second front orientation **N3** by inverting the banknotes **S** twice with the non-inverting unit **91**, while the banknote **S** in the second back orientation **N4** can be discharged from the inverting unit **92** in a state where the front-back directionality has been changed to the second front orientation **N3** by one front-back inversion with the inverting unit **92**.

(275) As a result, the front-back directionalities of the banknotes **S** discharged from the non-inverting unit **91** and the inverting unit **92** can be unified to the second front orientation **N3** by using the second front-back inverting unit **82**. Therefore, the banknotes **S** whose front-back directionality has been unified to the second front orientation **N3** can be transferred to the second accumulation and storage unit **101**.

(276) Here, in storing the banknotes **S** in the first accumulation and storage unit **100** and the second

accumulation and storage unit **101**, the control unit **51** preliminarily sets the spacing between the first base unit **140** and the second base unit **180** and the spacing between the third base unit **250** and the fourth base unit **400** to a spacing corresponding to the banknotes S.

(277) Specifically, the control unit **51**, upon recognizing that the denomination of the banknote S loaded based on the identification information of the identification unit **30** is a preset 10,000 yen note, reads the horizontal width direction and the vertical width direction of the 10,000-yen note from the data stored in the storage unit **51a**. The control unit **51**, on the basis of the read data, controls the first base unit **140** and the second base unit **180** to move along the front-rear direction **L3** so that the spacing in the front-rear direction **L3** between the first planar contact unit **162** and the second planar contact unit **202** matches the horizontal width of the 10,000-yen banknotes S to be accumulated on the upper base unit **112**, when the first alignment unit **160** and the second alignment unit **200** have been positioned in the first alignment position **K1** and second alignment position **K2**, as shown in FIG. **11**.

(278) In addition, the control unit **51** moves the first base unit **140** and the second base unit **180** along the front-rear direction **L3** so that the first planar contact unit **162** and the second planar contact unit **202** are arranged to be equally spaced from the reference position **O** in the upper base **112**, so that the reference position **O** in the upper base unit **112** and center of the banknote S in the horizontal width direction coincide.

(279) Further, the control unit **51**, according to the denomination of the banknote S preset by the operation display unit **50**, moves the third base unit **250** and the fourth base unit **400** along the left-right direction **L1**, to adjust the spacing between the third stopper unit **270** and the fourth stopper unit **420**. Thereby, as shown in FIG. **14**, according to the vertical width of the 10,000-yen banknotes S accumulated on the upper base unit **112**, the third stopper unit **270** and the fourth stopper unit **420** can be positioned facing each other in the left-right direction **L1** by the optimal spacing.

(280) A description will be given of how banknotes S discharged from the first front-back inverting unit **81** are stored in the first accumulation and storage unit **100** based on the initial settings described above.

(281) For example, as shown in FIG. **8**, when banknotes S whose front-back directionalities have been unified to the first front orientation **N1** are transferred from the non-inverting unit **91** of the first front-back inverting unit **81** to the first accumulation and storage unit **100**, the control unit **51** drives the third conveyance drive motor **302** to rotate the third conveyance shaft **284**. As a result, the third conveyance rollers **282** and the third impellers **283** can be rotated, and banknotes S can be conveyed toward the upper base unit **112** through the space between the upper surface of the third lower conveyance guide **281b** and the lower surface of the third upper conveyance guide **281a**. Therefore, the banknotes S can be loaded onto the upper base unit **112**.

(282) Similarly, when banknotes S whose front-back directionalities have been unified to the first front orientation **N1** are transferred from the inverting unit **92** to the first accumulation and storage unit **100**, the control unit **51** drives the fourth conveyance drive motor to rotate the fourth conveyance shaft **434**. As a result, the fourth conveyance rollers **432** and the fourth impellers **433** can be rotated, and banknotes S can be conveyed toward the upper base unit **112** through the space between the upper surface of the fourth lower conveyance guide **431b** and the lower surface of the fourth upper conveyance guide **431a**. Therefore, the banknotes S can be loaded onto the upper base unit **112**.

(283) In this way, the banknotes S discharged from the non-inverting unit **91** and the banknotes S discharged from the inverting unit **92** can for example be alternately loaded onto the upper base unit **112**, and can be stored while being accumulated on the upper base unit **112**.

(284) When the banknotes S are stored on the upper base unit **112**, the control unit **51** rotates the first alignment unit **160** and the second alignment unit **200** from the first retracted position **P1** and the second retracted position **P2** shown in FIG. **10** to the first alignment position **K1** and the second

alignment position **K2** shown in FIG. **11** almost simultaneously each time banknotes **S** are stored on the upper base unit **112** or each time a predetermined number of banknotes **S** are stored. As a result, the control unit **51** taps the banknotes **S** accumulated on the upper base unit **112** from both sides in the horizontal width direction. Thereby, the short sides of the banknotes **S** can be neatly unified, and the banknotes **S** can be aligned in the horizontal width direction.

(285) Moreover, since the first planar contact unit **162** and the second planar contact unit **202** are arranged at equal intervals from the reference position **O** on the upper base unit **112**, the banknotes **S** can be aligned so that the center of the banknotes **S** in the horizontal width direction matches the center of the banknotes **S** in the front-rear direction **L3** of the upper base unit **112** (that is, the center in the conveyance direction).

(286) Furthermore, since the control unit **51** drives the third conveyance drive motor **302**, the alignment plate drive shaft **331** can be rotated in conjunction with the rotation of the third conveyance rollers **282**. As a result, the eccentric cam **340** can be rotated together with the alignment plate drive shaft **331**, and the alignment plate **311** can be swung from the third retracted position **P3** shown in FIG. **16** to the third alignment position **K3** shown in FIG. **15**. Therefore, the banknotes **S** accumulated on the upper base unit **112** can be tapped from one side in the vertical width direction. As a result, as shown in FIG. **14**, the long sides of the banknotes **S** can be neatly unified, and the banknotes **S** can be aligned in the vertical width direction with respect to the fourth stopper unit **420**.

(287) Note that the alignment plate **311** may align the banknotes **S** each time they are accumulated on the upper base unit **112**, or may perform the alignment once every multiple number of sheets, such as once every two sheets, for example.

(288) As described above, banknotes **S** (10,000 yen notes) whose front-back directionality are unified in the first front orientation **N1** can be stored in the first accumulation and storage unit **100** while being neatly accumulated in a state of being aligned in the horizontal and vertical width directions.

(289) As in the case described above, by transferring the banknotes **S** from the second front-back inverting unit **82** into the second accumulation and storage unit **101**, banknotes **S** (10,000 yen notes) whose front-back directionality are unified in the second front orientation **N3** can be stored in the second accumulation and storage unit **101** while being neatly accumulated in a state of being aligned in the horizontal and vertical width directions.

(290) Next, when a predetermined number (for example, 100) of banknotes **S** are accumulated in the first accumulation and storage unit **100** and the second accumulation and storage unit **101**, as shown in FIG. **2**, the control unit **51** controls the accumulation conveyance unit **104** to remove the banknotes **S** in the accumulated state from the first accumulation and storage unit **100** and the second accumulation and storage unit **101** respectively, and convey them to the bundling unit **103**.

(291) Specifically, the control unit **51** drives the stopper drive motors in the first accumulation and storage unit **100** and the second accumulation and storage unit **101** to move the fourth stopper units **420** from the closed position **P4** (see FIG. **14**) to the open position. As a result, the banknote discharge port **404** can be opened, and the accumulated banknotes **S** can be taken out.

(292) Next, the control unit **51** moves the first chuck member **500** of the accumulation conveyance unit **104** toward the first accumulation and storage unit **100** side, and after removing the banknotes in an accumulated state stored on the upper base unit **112** of the first accumulation and storage unit **100**, transfers the banknotes **S** to the second chuck member **510** in the bundling unit **103**.

(293) Subsequently, the control unit **51** controls the bundling unit **103** to have the accumulated banknotes **S** received by the second chuck member **510** bundled by the bundling mechanism unit **520**. As a result, a small bundle in which the banknotes **S** in the accumulated state are bundled with the bundling tape can be created.

(294) Similarly, the control unit **51** moves the first chuck member **500** of the accumulation conveyance unit **104** toward the second accumulation and storage unit **101** side, and after removing

the banknotes in an accumulated state stored on the upper base unit **112** of the second accumulation and storage unit **101**, transfers the banknotes S to the second chuck member **510** in the bundling unit **103**.

(295) Subsequently, the control unit **51** controls the bundling unit **103** to have the accumulated banknotes S received by the second chuck member **510** bundled by the bundling mechanism unit **520**. As a result, a small bundle in which the banknotes S in the accumulated state are bundled with the bundling tape can be created.

(296) In this manner, the accumulated banknotes S stored in the first accumulation and storage unit **100** and the second accumulation and storage unit **101** can for example be taken out alternately and made into small bundles by the bundling unit **103**.

(297) The small bundles that are created are loaded into the small bundle discharge unit **521** shown in FIG. **1**. This allows the operator to take out and collect the small bundles from the small bundle discharge unit **521**. In particular, this small bundle is a bundle of banknotes S (10,000 yen notes) in which the front-back directionality of the banknotes S are unified and the banknotes S are neatly accumulated in the horizontal and vertical width directions.

(298) As described above, the banknote handling apparatus **1** of the present embodiment, using the first front-back inverting unit **81** and the second front-back inverting unit **82**, can perform front-back inversion of the banknotes S being conveyed by the coupled conveyance unit **70** based on the front-back identification information. Thereby, even when banknotes S are conveyed in a state in which the front-back directionality is mixed, the front-back directionality can be unified during conveyance.

(299) In particular, since processing can be performed that unifies the banknotes S so that the front and back faces thereof are facing the same way simply by providing a front-back inverting unit **80** (first front-back inverting unit **81** and the second front-back inverting unit **82**) equipped with the non-inverting unit **91** and the inverting unit **92** in the device case **60**, the configuration can be simplified and the entire banknote handling apparatus **1** can be made smaller and more compact.

(300) In particular, when the banknotes S have been conveyed to the first accumulation and storage unit **100** and the second accumulation and storage unit **101**, the banknote handling apparatus **1** of the present embodiment, by rotating the first alignment unit **160** and the second alignment unit **200** from the first retracted position P1 and the second retracted position P2 to the first alignment position K1 and the second alignment position K2, respectively, can tap the banknotes S accumulated on the upper base unit **112** from both sides in the horizontal width direction, and align the banknotes S in the horizontal width direction.

(301) In this way, by aligning the banknotes S by tapping from both sides in the horizontal width direction instead of aligning by tapping only from one side in the horizontal width direction as in the prior art, it is possible to reduce the moment stroke of the banknotes S (shift amount) and possible to efficiently perform reliable alignment. Therefore, the banknotes S can be quickly and reliably aligned, and the operation time required for the alignment process can be shortened.

(302) Furthermore, by positioning the first alignment unit **160** and the second alignment unit **200** at the first retracted position P1 and the second retracted position P2, the first alignment unit **160** and the second alignment unit **200** can be spaced outward of the entrance area of the banknotes S through which the banknotes S pass until being accumulated on the upper base unit **112**. Accordingly, it is possible to prevent the occurrence of a paper jam due to jamming of the banknotes S.

(303) Furthermore, since the first alignment unit **160** and the second alignment unit **200** have the first planar contact unit **162** and the second planar contact unit **202** that make linear or surface contact with the banknotes S, compared to the case of using a conventional cylindrical abutting portion, it is more difficult to leave a strike mark on the banknotes S. Moreover, when the banknotes S are tapped from both sides by using the conventional cylindrical abutting portions, depending on the position of the strike, there is a risk that the banknotes S may be rotated,

disrupting their alignment.

(304) In contrast, in the present embodiment, since the first planar contact unit **162** and the second planar contact unit **202**, which have wide and flat contact surfaces, are used it is possible to prevent the conventional rotation of the banknotes S that may occur and the accompanying disruption of the alignment of the banknotes S, and can stably align the banknotes S.

(305) Moreover, when the banknotes S have been conveyed to the first accumulation and storage unit **100** and the second accumulation and storage unit **101**, the banknote handling apparatus **1** of the present embodiment, by swinging the alignment plate **311** from the third retracted position P3 to the third alignment position K3, can tap the banknotes S accumulated on the upper base unit **112** from one side in the vertical width direction, and align the banknotes S in the vertical width direction with the fourth stopper unit **420**. As a result, due to the synergistic effect of the alignment in the horizontal width direction by the first alignment unit **160** and the second alignment unit **200** described above, the banknotes S can be accurately aligned both in the horizontal width direction and in the vertical width direction, and the banknotes S can be stored in an accumulated state that is neatly aligned.

(306) [Modification]

(307) In the present embodiment, when the banknotes S have been aligned in the horizontal width direction in the first accumulation and storage unit **100** and the second accumulation and storage unit **101**, the first alignment unit **160** and the second alignment unit **200** were nearly simultaneously rotated from the first retracted position P1 and the second retracted position P2 to the first alignment position K1 and the second alignment position K2, respectively, but the present invention is not limited to this case, and the first alignment unit **160** and the second alignment unit **200** may be rotated at different timings.

(308) For example, the control unit **51** obtains banknote conveyance position information indicating the position of the banknote S being conveyed with respect to the horizontal width direction from the identification unit **30**, and, as shown in FIG. **18**, if the center position C in the horizontal width direction of the banknotes S accumulated on the upper base unit **112** is closer to one side (to the first alignment unit **160** side) than the reference position O by the upper base unit **112**, as shown in FIG. **18**, the second alignment unit **200** is rotated first to the second alignment position K2. As a result, the second planar contact unit **202** can be kept on standby so as to be substantially parallel to the vertical width direction of the banknotes S, which is the conveying direction.

(309) Subsequently, as shown in FIG. **20**, the first alignment unit **160** is rotated to the first alignment position K1 to tap the banknotes S, pushing the banknotes S so as to abut against the second planar contact unit **202** of the second alignment unit **200**, which is already standing by in the second alignment position K2. Thereby, the banknotes S accumulated on the upper base unit **112** can be aligned in the horizontal width direction.

(310) Contrary to the case described above, when the center position C in the horizontal width direction of the banknotes S accumulated on the upper base unit **112** is shifted to the other side (the second alignment unit **200** side) of the reference position O by the upper base unit **112**, the first alignment unit **160** is first rotated to the first alignment position K1. As a result, the first planar contact unit **162** can be kept on standby so as to be substantially parallel to the vertical width direction of the banknotes S, which is the conveying direction. Subsequently, the second alignment unit **200** is rotated to the second alignment position K2 to tap the banknotes S, drawing the banknotes S together so as to abut against the first planar contact unit **162** of the first alignment unit **160**, which is already waiting at the first alignment position K1. Thereby, the banknotes S accumulated on the upper base unit **112** can be aligned in the horizontal width direction.

(311) Therefore, even in the case described above, the banknotes S accumulated on the upper base unit **112** can be neatly aligned in the horizontal width direction.

(312) Although embodiments of the present invention have been described above, these

embodiments are presented as examples and are not intended to limit the scope of the invention. Embodiments can be implemented in various other forms, and various omissions, replacements, and modifications can be made without departing from the scope of the invention. Embodiments and modifications thereof include, for example, those that can be easily imagined by those skilled in the art, those that are substantially the same, and those within an equivalent range.

(313) For example, in the above-described embodiment, the case in which the banknote handling apparatus **1** is equipped with the identification and counting device **2** and the accumulation and bundling device **3** that are coupled to each other was described as an example, but the identification and counting device **2** is not essential and does not have to be provided.

(314) Furthermore, the configuration is not limited to the case where only one accumulation and bundling device **3** is provided, and two or more may be provided.

(315) For example, the banknote handling apparatus **1** may be formed by coupling two accumulation and bundling devices **3** in the left-right direction **L1**. In this case, by connecting the coupled conveyance unit **70** in the second accumulation and bundling device **3** to the option outlet **71** of the coupled conveyance unit **70** in the first accumulation and bundling device **3**, the banknotes **S** can be continuously conveyed from the first accumulation and bundling device **3** to the second accumulation and bundling device **3**.

(316) When the banknote handling apparatus **1** is configured in this way, for example, the soiled banknotes **S** identified as soiled banknotes **S** (soiled paper sheets) by the identification unit **30** can be passed through the first accumulation and bundling device **3**, and small bundles of soiled banknotes **S** can be made in the second accumulation and bundling device **3**, while unifying the front-back directionality thereof. Therefore, the second accumulation and bundling device **3** can be used as a dedicated device for bundling the damaged banknotes **S**.

(317) This allows for uses such as making small bundles while separating the soiled banknotes **S** from the other banknotes **S**.

(318) Furthermore, instead of soiled banknotes **S**, utilizing a second accumulation and bundling device **3** allows for uses such as making small bundles while unifying the front-back directionality of banknotes **S** of denominations different from those of the first accumulation and bundling device **3**. For example, the first accumulation and bundling device **3** can be used for banknotes **S** of 10,000 yen as in the above embodiment, while the second accumulation and bundling device **3** can be used for banknotes **S** of 5,000 yen.

(319) Furthermore, a total of four accumulation and bundling devices **3** can be coupled for use as a dedicated accumulation and bundling device **3** for banknotes **S** of 10,000-yen notes, 5,000-yen notes, 2,000-yen notes, and 1,000-yen notes.

(320) Furthermore, the above embodiment is configured with the first front-back inverting unit **81** and the second front-back inverting unit **82**, but the first front-back inverting unit **81** and the second front-back inverting unit **82** are not essential and do not have to be provided.

(321) Even in this case, for example, by conveying the banknotes **S** with their front-back directionalities unified to the coupled conveyance unit **70**, the first accumulation and storage unit **100** and the second accumulation and storage unit **101** can be used to store the banknotes **S** in an accumulated state while aligning them.

(322) Furthermore, in the case where the first front-back inverting unit **81** and the second front-back inverting unit **82** are not provided, for example, the first accumulation and storage unit **100** and the second accumulation and storage unit **101** may be arranged side by side in the vertical direction **L2**.

(323) Specifically, as shown in FIG. **21**, the second accumulation and storage unit **101** may be arranged below the first accumulation and storage unit **100**. In this case, a branch conveyance unit **600** branched from the coupled conveyance unit **70** may be connected to the first accumulation and storage unit **100**, and a second branch conveyance unit **601** further branched from the first branch conveyance unit **600** may be connected to the second accumulation and storage unit **101**. Further,

the bundling unit **103** can be arranged below the first accumulation and storage unit **100** and the second accumulation and storage unit **101**, and the accumulation conveyance unit **104** can be arranged so as to move in the vertical direction **L2** between the first accumulation and storage unit **100**, the second accumulation and storage unit **101**, and the bundling unit **103**.

(324) Even in this configuration, it is possible to create small bundles by bundling with the bundling unit **103** banknotes **S** in an aligned and accumulated state in the first accumulation and storage unit **100** and banknotes **S** in an aligned and accumulated state in the second accumulation and storage unit **101**.

(325) Furthermore, in the above-described embodiment, Japanese banknotes **S** are used as an example of paper sheets, but as described above, foreign banknotes **S** may be used, and the embodiment is not limited to banknotes **S**. For example, the present invention can be applied to a device that processes paper sheets in general, such as gift certificates, checks, money market instruments such as commercial paper, and securities.

(326) In particular, according to the first accumulation and storage unit **100** and the second accumulation and storage unit **101** of the present embodiment, since according to the size of the paper sheets the spacing between the first base unit **140** and the second base unit **180** can be adjusted, and the spacing between the third base unit **250** and the fourth base unit **400** can be adjusted, it is possible to accommodate a variety of paper sheets, thereby enabling an enhancement of convenience and ease of use.

(327) Furthermore, in the above-described embodiment, when setting the banknotes **S** in the loading unit **11**, they are set in an orientation aligning the long sides of the banknotes **S**, that is, the horizontal width direction, with the front-rear direction **L3** of the apparatus, and aligning the short sides of the banknotes **S**, that is, the vertical width direction, with the left-right direction **L1** of the apparatus, but the embodiment is not limited to this case, and the apparatus may be constituted so that the banknotes **S** are set in an orientation aligning the long sides of the banknotes **S**, that is, the horizontal width direction, with the left-right direction **L3** of the apparatus, and aligning the short sides of the banknotes **S**, that is, the vertical width direction, with the front-rear direction **L3**.

(328) Furthermore, in the above embodiment, although the description was given by taking the example of the configuration that applies the first alignment unit **160** and the second alignment unit **200** to the width-direction alignment unit **120** in the first accumulation and storage unit **100** and the second accumulation and storage unit **101** to align the banknotes in the horizontal width direction, the invention is not limited to this case. That is, the first alignment unit **160** and the second alignment unit **200** may be applied to the conveyance direction alignment unit **130** to align the banknotes **S** in the vertical width direction. In other words, the first alignment unit and the second alignment unit according to the present invention can be used in either case of aligning the banknotes **S** in the horizontal width direction or in the case of aligning the banknotes **S** in the vertical width direction.

INDUSTRIAL APPLICABILITY

(329) The present invention can be applied to a paper sheet handling apparatus for handling paper sheets such as banknotes, and can provide a paper sheet handling apparatus capable of quickly and reliably aligning paper sheets.

REFERENCE SIGNS LIST

(330) **C** Center position **O** Reference position **S** Banknote (paper sheet) **K1** First alignment position **K2** Second alignment position **K3** Third alignment position **P1** First retracted position **P2** Second retracted position **P3** Third retracted position **1** Banknote handling apparatus (paper sheet handling apparatus) **2** Identification and counting device **11** Loading unit **20** Identification and conveyance unit **30** Identification unit **40** Outbound conveyance unit **51** Control unit **70** Coupled conveyance unit (conveyance unit) **100** First accumulation and storage unit (accumulation and storage unit) **101** Second accumulation and storage unit (accumulation and storage unit) **104** Accumulation conveyance unit **110** Unit base unit (accumulation unit) **140** First base unit **160** First alignment unit

162 First planar contact unit **180** Second base unit **200** Second alignment unit **202** Second planar contact unit **250** Third base unit **280** Third conveyance unit (Third accumulation conveyance unit) **282** Third conveyance roller **311** Alignment plate **380** Alignment plate drive unit **400** Fourth base unit **430** Fourth conveyance unit (Fourth accumulation conveyance unit) **432** Fourth conveyance roller

Claims

1. A paper sheet handling apparatus comprising: a conveyance unit that conveys paper sheets; an accumulation and storage unit that accumulates and stores paper sheets conveyed by the conveyance unit while aligning the paper sheets; and a control unit that controls the accumulation and storage unit, wherein the accumulation and storage unit comprises: an accumulation unit that accumulates paper sheets so that a first direction, which is one direction of the vertical width direction and the horizontal width direction, and a second direction, which is the other of the vertical width direction and the horizontal width direction, are oriented in predetermined directions; a first alignment unit that is arranged more to one side in the first direction than the accumulation unit and constituted to be movable between a first retracted position spaced apart from the paper sheets accumulated in the accumulation unit, and a first alignment position that is in contact with one end edge positioned in the first direction of the paper sheets accumulated in the accumulation unit; and a second alignment unit that is arranged more to the other side in the first direction than the accumulation unit and constituted to be movable between a second retracted position spaced apart from the paper sheets accumulated in the accumulation unit, and a second alignment position that is in contact with the other end edge positioned in the first direction of the paper sheets accumulated in the accumulation unit, a third base unit disposed more to one side in the second direction than the accumulation unit and configured to be movable along the second direction; a fourth base unit disposed more to the other side in the second direction than the accumulation unit, configured to be movable along the second direction, and configured to be capable of approaching and separating from the third base unit; a third accumulation conveyance unit mounted on the third base unit and having a third conveyance roller for loading the paper sheets conveyed from the conveyance unit into the accumulation unit; and a fourth accumulation conveyance unit mounted on the fourth base unit and having a fourth conveyance roller for loading the paper sheets conveyed from the conveyance unit into the accumulation unit, wherein the control unit causes the paper sheets to be aligned in the first direction by causing the first alignment unit and the second alignment unit to move from the first retracted position and the second retracted position to the first alignment position and the second alignment position and tap the paper sheets accumulated in the accumulation unit from both sides in the first direction, and further causes the third base unit and the fourth base unit to move along the second direction according to the type of paper sheets accumulated in the accumulation unit, to adjust the spacing between the third base unit and the fourth base unit.

2. The paper sheet handling apparatus according to claim 1, wherein the accumulation and storage unit further comprises: a first base unit disposed more to one side in the first direction than the accumulation unit and configured to be movable along the first direction; and a second base unit disposed more to the other side in the first direction than the accumulation unit, provided to be movable along the first direction, and configured to be capable of approaching and separating from the first base unit, wherein the first alignment unit is mounted on the first base unit, the second alignment unit is mounted on the second base unit, and the control unit causes the first base unit and the second base unit to move along the first direction according to the type of paper sheets accumulated in the accumulation unit, to adjust the spacing between the first alignment unit and the second alignment unit.

3. The paper sheet handling apparatus according to claim 2, wherein the first alignment unit has a

first planar contact unit that is arranged along the second direction of the paper sheets and contacts the one end edge of the paper sheets when positioned at the first alignment position, the second alignment unit has a second planar contact unit that is arranged along the second direction of the paper sheets and contacts the other end edge of the paper sheets when positioned at the second alignment position, and the control unit causes the first base unit and the second base unit to move along the first direction so that, when the first alignment unit and the second alignment unit are positioned at the first alignment position and the second alignment position, the spacing in the first direction between the first planar contact unit and the second planar contact unit matches the length in the first direction of the paper sheets accumulated in the accumulation unit.

4. The paper sheet handling apparatus according to claim 1, wherein the control unit causes the first alignment unit and the second alignment unit to move simultaneously from the first retracted position and the second retracted position to be positioned at the first alignment position and the second alignment position.

5. The paper sheet handling apparatus according to claim 1, wherein, when the paper sheets accumulated in the accumulation unit are accumulated in a state of the center position of the paper sheets in the first direction being shifted closer to the first alignment unit than a reference position located at substantially a center of the first direction and the second direction of the accumulation unit of the accumulation unit, the control unit aligns the paper sheets by, after first causing the second alignment unit to move to the second alignment position, causing the first alignment unit to tap the paper sheets by moving to the first alignment position, thereby causing the paper sheets to abut the second alignment unit standing by at the second alignment position, and wherein, when the paper sheets accumulated in the accumulation unit are accumulated in a state of the center position being shifted closer to the second alignment unit than the reference position, the control unit aligns the paper sheets by, after first causing the first alignment unit to move to the first alignment position, causing the second alignment unit to tap the paper sheets by moving to the second alignment position, thereby causing the paper sheets to abut the first alignment unit standing by at the first alignment position.

6. The paper sheet handling apparatus according to claim 1, further comprising: an alignment plate that is mounted on the third base unit and is configured to be movable between a third retracted position separated from the paper sheets accumulated in the accumulation unit, and a third alignment position that is in contact with one end edge positioned in the second direction of the paper sheets accumulated in the accumulation unit; and an alignment plate drive unit that moves the alignment plate from the third retracted position to the third alignment position in conjunction with the third conveyance roller, wherein the control unit aligns the paper sheets accumulated in the accumulation unit in the second direction by driving the alignment plate drive unit in accordance with the type of paper sheets accumulated in the accumulation unit to move the alignment plate from the third retracted position to the third alignment position.
